

Supplemental Material for: Spectral diagnostics and local fixed-budget sensitivity of critical points in finite encounter-reaction models

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This Supplemental Material is a standalone detailed record for the associated article. It gives the full coordinate and reaction-model qualifications, finite-support Green and spectral audits, generalized-inverse-Gaussian screening construction, finite-state discovery and clock controls, complete two-dimensional discretization and endpoint families, capacity calibrations, multichannel screening calculations, normal-form derivations, provenance boundary, and limitations. The main article retains only the Green–Woodbury reduction, the necessary spectral gate, fixed-budget sensitivity, the finite CTMC fold, and the M2D-F finite-generator fold certificate. All results here remain finite-model or screening statements unless an exact identity is explicitly identified; no continuum modality theorem is asserted.

S1. SCOPE AND DETAILED EVIDENCE MAP

This file is the technical companion to the focused main article, not a second statement of its narrative. It retains the full model qualifications, derivations, secondary finite-grid families, screening calculations, provenance boundary, and limitations needed to audit the evidence. The present work is a direct extension of that encounter and heterogeneous-transmission program; it does not claim a new Green, multi-target, fold-criterion, or fixed-total-reactivity theory. The finite evidence uses separately declared M2D-E, M2D-F, M2D-C, and M2D-T families. No common continuation relation between M2D-E and M2D-F is asserted.

Three evidence levels remain separate throughout: exact identities under stated hypotheses, certificates for declared finite generators, and numerical screening or continuum-consistency evidence. In particular, the finite-grid folds and endpoint audits do not establish a continuum modality theorem.

Table SI fixes the evidence level and corresponding claim boundary for every analytical and numerical layer used below.

The spatial-channel schematic and generic fold geometry appear as Fig. 1 of the main article. The supplemental record therefore begins directly with the model qualifications and derivations rather than duplicating that figure.

S2. FINITE-RADIUS PATTERNED ENCOUNTER MODEL

S2.1. Joint dynamics and reaction channels

Let $X_i(t) \in \Omega \subset \mathbb{R}^d$ denote two independent particles before reaction,

$$dX_i = b_i(X_i) dt + \sqrt{2D_i} dW_i, \quad D_i > 0, \quad i = 1, 2, \quad (\text{S1})$$

with independent Brownian motions. We use reflecting outer boundaries unless stated otherwise. The unreacted joint density $p(x_1, x_2, t)$ satisfies

$$\partial_t p = \mathcal{L}_0 p - K(x_1, x_2)p, \quad (\text{S2})$$

$$\mathcal{L}_0 p = -\nabla_{x_1} \cdot (b_1 p) - \nabla_{x_2} \cdot (b_2 p) + D_1 \Delta_{x_1} p + D_2 \Delta_{x_2} p, \quad (\text{S3})$$

with no-flux currents $J_i = b_i p - D_i \nabla_{x_i} p$.

The finite contact radius is $a > 0$. A physical catalyst must also declare which affine location of the pair it reads:

$$C_\eta(x_1, x_2) = \eta x_1 + (1 - \eta)x_2, \quad 0 \leq \eta \leq 1. \quad (\text{S4})$$

Catalytic patch C_j is specified in this C_η -space, with intrinsic Doi rate $\kappa_j(C_\eta) \geq 0$. The channel-resolved killing fields are

$$K_{a,j}^{(\eta)}(x_1, x_2) = \kappa_j(C_\eta) \mathbf{1}_{C_j}(C_\eta) \mathbf{1}_{\{|x_1 - x_2| < a\}},$$

$$K_a^{(\eta)} = \sum_{j=1}^m K_{a,j}^{(\eta)}. \quad (\text{S5})$$

The survival probability and reaction-channel densities are

$$S(t) = \int_{\Omega^2} p(x_1, x_2, t) dx_1 dx_2, \quad (\text{S6})$$

$$f_j(t) = \int_{\Omega^2} K_{a,j}^{(\eta)}(x_1, x_2) p(x_1, x_2, t) dx_1 dx_2, \quad (\text{S7})$$

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TABLE SI. Statement/evidence boundary used in this manuscript.

Statement	Evidence level	Boundary
Coordinate decoupling, mass balance, Woodbury reduction, fixed-shape fold, generic normal form	Exact identities under stated algebraic or operator hypotheses	Inverse-free volume-reaction identity; no continuum meromorphic continuation, bounded-coordinate independence, or unqualified Robin trace claim
Finite reversible spectral diagnostic	Classical generalized-Descartes corollary plus spectral reconstructions of the 961-state encounter and 2025-state M2D-T primary generators	Necessary sign-variation bound only; not sufficient, not a channel-rank bound, and not applied to complex-spectrum, Jordan, or continuum laws; this is a loose consistency/falsification gate, not independent root discovery
Fixed-budget modality susceptibility	Exact finite Fréchet–Duhamel identity, closed-form projected optimum, 256 independent directions, and 20,000 random feasible controls	Infinitesimal interior-rate design only; no finite-amplitude, binary-mask, moving-support, or continuum optimum is claimed
Finite CTMC physical fold	Finite-matrix derivative identities evaluated numerically, plus a root solve and independent continuation	One finite model; minority splitting probability 8.95×10^{-6}
M2D-F matched-budget folds	Three nondegenerate finite-generator folds and a budget-measure control	Separate finite-model mechanism certificates; fold coordinates span 0.262 and move under product-control-volume weighting; no common continuation with M2D-E or continuum critical value is claimed
M2D-E equal-budget endpoint amplification	A five-grid resolution, tail, morphology, and 101-cutoff audit	Four finer patterned endpoints are resolved-bimodal and the coarse endpoint is a shoulder; the cutoff interval is post-hoc and sample-derived, not a detector/noise-robustness result; no M2D-E fold is claimed
2D catalytic-coordinate sensitivity	One-factor midpoint/weighted comparison on three principal grids	Labels agree but masks and densities change; the physical sink models are not equivalent
Bounded 2D three-channel trimodality	Five detected sign-changing derivative roots of alternating type (hence at least five positive-time roots) and ordered channel attribution on four finite grids; three classifier-resolved trimodal cases and one three-maximum shoulder	Patch radii are below h_x , reactive-state counts are nonmonotone, and there is no interval-exhaustive root certificate, cell-averaged continuum trimodal region, or phase boundary
2D and 3D capacity laws	Translation-invariant relative-coordinate mean-time calculations	Not centre-patterned density or modality calculations
Multidimensional multi-channel GIG construction	Exact design algebra plus floating-point derivative-root isolation in 12 declared screening cases	Not a bounded Doi three- or four-mode theorem
GIG encounter channels	Exact GIG algebra, approximate mapping from free encounter geometry	No uniform confined-channel remainder

Reflecting outer boundaries give the exact mass balance

$$-S'(t) = f(t) = \sum_{j=1}^m f_j(t). \quad (\text{S8})$$

The splitting probability into channel j is $\pi_j = \int_0^\infty f_j(t) dt$; when reaction is certain, $\sum_j \pi_j = 1$, and $g_j = f_j/\pi_j$ is the conditional channel-time density.

S2.2. Relative, diffusion-decoupling, and catalytic coordinates

Set

$$D_r = D_1 + D_2, \quad D_c = \frac{D_1 D_2}{D_1 + D_2}, \quad (\text{S9})$$

and define

$$r = X_1 - X_2, \quad R = \frac{D_2 X_1 + D_1 X_2}{D_1 + D_2}. \quad (\text{S10})$$

The catalytic coordinate in Eq. (S4) is related to the diffusion-decoupling coordinate by

$$C_\eta = R + \left(\eta - \frac{D_2}{D_r} \right) r. \quad (\text{S11})$$

Hence $C_{D_2/D_r} = R$. The bounded two-dimensional calculations below use the physical midpoint convention

$$C_{1/2} = \frac{X_1 + X_2}{2} = R + \frac{D_1 - D_2}{2D_r} r, \\ |C_{1/2} - R| \leq \frac{|D_1 - D_2|}{2D_r} a, \quad |r| < a. \quad (\text{S12})$$

The inverse map is

$$x_1 = R + \frac{D_1}{D_r} r, \quad x_2 = R - \frac{D_2}{D_r} r, \quad (\text{S13})$$

and the absolute Jacobian is one. Direct substitution gives the exact cancellation

$$D_1 \Delta_{x_1} + D_2 \Delta_{x_2} = D_r \Delta_r + D_c \Delta_R. \quad (\text{S14})$$

Thus

$$\partial_t q = -\nabla_r \cdot (uq) - \nabla_R \cdot (v_c q) + D_r \Delta_r q + D_c \Delta_R q \\ - K_a^{(\eta)}(x_1(R, r), x_2(R, r))q, \quad (\text{S15})$$

where

$$u = b_1(x_1) - b_2(x_2), \quad v_c = \frac{D_2 b_1(x_1) + D_1 b_2(x_2)}{D_r}. \quad (\text{S16})$$

For constant drifts in free space, relative and centre noises are independent. This statement must not be transferred unqualified to bounded domains: the transformed domain

$$\mathcal{D} = \left\{ (R, r) : R + \frac{D_1}{D_r} r \in \Omega, \quad R - \frac{D_2}{D_r} r \in \Omega \right\} \quad (\text{S17})$$

has coupled boundaries. Nonconstant drifts can also couple r and R . The weighted coordinate in Eq. (S10) is selected by diffusion, not imposed as the physical catalyst location. In (r, C_η) coordinates, the centre diffusivity is $D_\eta = D_1 \eta^2 + D_2(1 - \eta)^2$ and the mixed second-order coefficient is $2[\eta D_1 - (1 - \eta)D_2]$. It vanishes only for $\eta = D_2/D_r$. For $\eta = 1/2$ the mixed coefficient is $D_1 - D_2$; equivalently, the decoupled (r, R) representation retains the slanted sink support in Eq. (S11). The Green formulation below accepts either declared joint-space sink, whereas the factorized GIG screening law uses the special choice $C_\eta = R$.

S2.3. Doi and radiation descriptions

The dimensionless Doi strength is

$$\text{Da}_{\text{Doi}} = \frac{\kappa a^2}{D_r}. \quad (\text{S18})$$

Consequently, a radius comparison at fixed intrinsic regime scales $\kappa \propto a^{-2}$. Holding a per-grid-site reaction probability fixed does not define a continuum comparison in $d \geq 2$.

One can alternatively remove $|r| < a$ and impose a radiation condition on the contact surface. With outward normal $n_{\mathcal{D}}$ for the admissible domain $|r| > a$,

$$J_r \cdot n_{\mathcal{D}} = \sum_j \kappa_{s,j}(C_\eta, \omega) \mathbf{1}_{C_j}(C_\eta) q, \quad |r| = a, \quad (\text{S19})$$

and $\text{Da}_{\text{rad}} = \kappa_s a / D_r$. Doi and radiation models are not identical at equal bare parameters. They can be matched through effective reactivity in a small-target regime [1, 2]; no independent Robin calculation is claimed in the present numerical results.

S3. REACTION-SUPPORT GREEN REDUCTION

Let $\mathbf{H} = L^2(\mathcal{D})$. Write the Doi killing operator as

$$V = \Gamma^* \mathbf{K} \Gamma, \quad \mathcal{T} = \mathcal{L}_0 - V, \quad (\text{S20})$$

where Γ restricts a function to a declared volume-reaction support, Γ^* extends it by zero, and $\mathbf{K}$ is a bounded non-negative multiplication operator on that support. It may have a kernel: zero-rate channels and zero-rate subsets are permitted. For $\text{Re } s$ above the spectral bound, define

$$\mathcal{R}_0(s) = (s - \mathcal{L}_0)^{-1}, \quad \mathcal{G}(s) = \Gamma \mathcal{R}_0(s) \Gamma^*. \quad (\text{S21})$$

Put $\mathbf{M} = I + \mathcal{G}\mathbf{K}$. Whenever the displayed bounded operators and inverse exist, the inverse-free resolvent identity is

$$(s - \mathcal{T})^{-1} = \mathcal{R}_0 - \mathcal{R}_0 \Gamma^* \mathbf{K} \mathbf{M}^{-1} \Gamma \mathcal{R}_0. \quad (\text{S22})$$

If $u = \Gamma \mathcal{R}_0 q_0$, the restricted state and transformed reaction density on the support are

$$x = \Gamma (s - \mathcal{T})^{-1} q_0 = \mathbf{M}^{-1} u, \\ y = \mathbf{K} x = \mathbf{K} \mathbf{M}^{-1} u. \quad (\text{S23})$$

No inverse of $\mathbf{K}$ is required. If $\mathbf{K}$ is boundedly invertible on the chosen support, the push-through identity gives the optional corollary

$$\mathbf{K}(I + \mathcal{G}\mathbf{K})^{-1} = (\mathbf{K}^{-1} + \mathcal{G})^{-1}. \quad (\text{S24})$$

Equations (S22) and (S23) show precisely how free transport, reaction geometry, and intrinsic rates enter. For a volume sink they are conditional operator identities under the usual domain assumptions. A Robin analogue lives in trace spaces and cannot be obtained by treating boundary restriction as a bounded L^2 map without qualification.

S3.1. Finite hotspots and splitting probabilities

For a finite CTMC, use the row-vector convention. Let $L_0 = L_1 \oplus L_2$ be the free two-particle generator, let the columns of U select the catalytic co-location states, and let $K = \text{diag}(\kappa_1, \dots, \kappa_m)$. Then

$$T = L_0 - UKU^\top, \quad (\text{S25})$$

and for an initial distribution α ,

$$f_j(t) = \alpha e^{Tt} UK e_j. \quad (\text{S26})$$

The free hotspot Green matrix is

$$G(s) = U^\top (sI - L_0)^{-1} U. \quad (\text{S27})$$

For finite matrices and any $s \notin \sigma(L_0)$, define the inverse-free renewal determinant

$$\Delta(s) = \det[I + G(s)K]. \quad (\text{S28})$$

The matrix determinant lemma gives

$$\det(sI - T) = \det(sI - L_0) \Delta(s). \quad (\text{S29})$$

For two hotspots,

$$\Delta(s) = (1 + \kappa_1 g_{11})(1 + \kappa_2 g_{22}) - \kappa_1 \kappa_2 g_{12} g_{21}. \quad (\text{S30})$$

When both rates are nonzero, this is $\kappa_1 \kappa_2 \det(K^{-1} + G)$; the latter normalization is not used at a zero rate. A zero of Δ is only a candidate killed pole coupled to the reactive subspace. Numerator cancellation, vanishing observable residue, and modes dark to U must be checked separately. A root of $\Delta(s)$ is not, by itself, a time-domain mode of the density. Negative-half-plane evaluation here is finite rational matrix algebra, not a Laplace transform. For the continuum operator, the present Green result remains a right-half-plane Laplace-response identity; meromorphic continuation and continuum pole/residue claims require additional Fredholm hypotheses not established here.

An exact finite two-site audit exercises all three exclusions. With identical two-state walkers, rates $\kappa_1 = \kappa_2 = 1/2$, and co-location selectors, $v_{\text{dark}} = (0, 1, -1, 0)^\top$ obeys

$$U^\top v_{\text{dark}} = 0, \quad L_0 v_{\text{dark}} = T v_{\text{dark}} = -2v_{\text{dark}}. \quad (\text{S31})$$

At the coupled killed pole $s_* = -5/2 \notin \sigma(L_0)$,

$$I + G(s_*)K = \frac{8}{15} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad r_{\text{ch}} = \left(\frac{1}{4}, -\frac{1}{4} \right). \quad (\text{S32})$$

Thus both channels carry a nonzero residue while the ordinary total reaction flux cancels this pole exactly. This finite rational audit validates the warning; it is not a continuum spectral theorem and no modality conclusion is inferred from it.

The free reflecting generator has a zero mode, so $G(s)$ is singular as $s \downarrow 0$. Splitting probabilities are evaluated stably from the killed solve,

$$\pi = \alpha(-T)^{-1} UK, \quad (\text{S33})$$

rather than by substituting $s = 0$ into individually divergent free resolvents, provided the full killed live operator is transient and $-T$ is invertible. Initial-class-only transience requires restriction to the reachable transient subspace. At small nonzero s , the finite Green routine also fails closed if its combined solve, condition-number, reconstructed equation, or direct-comparison diagnostic exceeds 10^{-8} ; it never silently switches methods. The exact time derivatives are

$$\partial_t^n f(t) = \alpha e^{Tt} T^n UK \mathbf{1}. \quad (\text{S34})$$

For a parameter θ ,

$$\partial_\theta(s - T)^{-1} = (s - T)^{-1} T_\theta (s - T)^{-1}, \quad (\text{S35})$$

Writing $R = (s - T)^{-1}$ and $F = \alpha R UK$, the complete fixed-space sensitivity is

$$\begin{aligned} \partial_\theta F &= \alpha_\theta R UK + \alpha R T_\theta R UK \\ &\quad + \alpha R U_\theta K + \alpha R U K_\theta. \end{aligned} \quad (\text{S36})$$

The fixed-initial-law, fixed-support rate Jacobian used below retains the second and fourth terms. Moving a sharp patch changes U and need not be a smooth parameter on a fixed grid. These identities avoid finite-difference derivatives at a modality fold.

S4. SPECTRAL CONSTRAINTS AND LOCAL FIXED-BUDGET SENSITIVITY

S4.1. A necessary spectral sign-variation bound

Consider first a finite killed model whose observable density has a real, diagonalizable spectral expansion. After equal decay rates are grouped and zero residues removed, write

$$f(t) = \sum_{j=1}^n a_j e^{-\lambda_j t}, \quad 0 < \lambda_1 < \dots < \lambda_n. \quad (\text{S37})$$

Reversibility is a sufficient, not necessary, hypothesis: detailed balance makes the killed generator similar to a real symmetric matrix, while the residues are products of the initial-state and reaction-observable spectral projections. Applying the classical generalized Descartes rule [3] to

$$f_t(t) = - \sum_{j=1}^n \lambda_j a_j e^{-\lambda_j t} \quad (\text{S38})$$

after the substitution $x = e^{-t}$ gives the following immediate corollary.

Spectral sign-variation corollary. The number of positive-time zeros of f_t , counted with multiplicity, is at most the number $V(a_1, \dots, a_n)$ of sign changes in the ordered residue sequence. Consequently, m nondegenerate interior maxima separated by minima require

$$V(a_1, \dots, a_n) \geq 2m - 1, \quad m \leq \left\lfloor \frac{V+1}{2} \right\rfloor. \quad (\text{S39})$$

Thus bimodality needs at least three sign changes and trimodality at least five. This is a classical zero-count consequence used here as a spectral design diagnostic, not a new theorem. It complements established shape results for phase-type densities [4]; it does not cover complex eigenvalues, Jordan terms, or an infinite continuum expansion.

The bound is deliberately one-sided. For example,

$$4e^{-t} - 12e^{-2t} + 12e^{-3t} - 4e^{-4t} = 4e^{-t}(1 - e^{-t})^3 \quad (\text{S40})$$

has three residue sign changes but only one positive critical point, $t = \log 4$. Nor can reaction-support rank replace the residue condition. Two 12-stage Erlang transport branches with rates 12 and 1.2, initialized with equal weight and feeding a single state killed at rate 100, have direct matrix-exponential critical points at 0.92676432 (maximum), 2.65921039 (minimum), and 9.17667747 (maximum). A rank-one killing support can therefore already be bimodal. Transport clocks and observable residues, not the number of patch labels, control the available temporal structure.

Full symmetric eigendecompositions test Eq. (S39) in the primary models. The 961-state supercritical finite encounter CTMC has three reported critical points, and the 2025-state 9×5 M2D-T model is evaluated at five detected critical points. The spectral audit rebuilds the same primary generators and imports or reconstructs those reported critical times; it is therefore not an independent root-finding implementation. Both cases retain more than the required three and five sign changes when grouped residues below relative cutoffs from 10^{-12} through 10^{-6} are removed. Density and derivative reconstructions are checked at every supplied root. The observed sign counts are much larger than the minima and are not interpreted as predicted mode counts: across the stated cutoff sweep, the minimum retained counts are 444 versus the required 3 in the bimodal case and 191 versus the required 5 in the trimodal case. This is a loose falsification gate, not a mode-count predictor.

S4.2. Budget-projected modality susceptibility

The spectral corollary can rule out a mode count but does not tell us how to move the reaction field. For a finite live-state transport generator L , let

$$T(k) = L - \text{diag } k, \quad f(t; k) = \alpha e^{T(k)t} k, \quad (\text{S41})$$

where $k > 0$ on the allowed design support. For an additive perturbation $k_\epsilon = k + \epsilon h$, put $H = -\text{diag } h$. The exact finite-matrix Fréchet–Duhamel identity is

$$De^{Tt}[H] = \int_0^t e^{T(t-s)} H e^{Ts} ds. \quad (\text{S42})$$

This is the standard Fréchet derivative of the matrix exponential [5]. Since $f_t = \alpha e^{Tt} T k$, its complete directional derivative is

$$D_k f_t(t; k)[h] = \alpha De^{Tt}[H] T k + \alpha e^{Tt} (H k + T h). \quad (\text{S43})$$

The second term is essential because the killing field is also the measured reaction observable. Equation (S43) is linear in h , so write it as $g^\top h$. Componentwise,

$$g_i = (\alpha e^{Tt} T)_i - k_i (\alpha e^{Tt})_i - \int_0^t (\alpha e^{T(t-s)})_i (e^{Ts} T k)_i ds. \quad (\text{S44})$$

Let $c > 0$ define the chosen discrete killing budget, $c^\top h = 0$, and let $M \succ 0$ define the local design norm $h^\top M h = 1$. Setting

$$\lambda_B = \frac{c^\top M^{-1} g}{c^\top M^{-1} c}, \quad \tilde{g} = g - \lambda_B c, \quad (\text{S45})$$

Cauchy–Schwarz on the budget tangent space gives

$$h_* = \frac{M^{-1} \tilde{g}}{\sqrt{\tilde{g}^\top M^{-1} \tilde{g}}}, \quad \max D_k f_t[h] = \sqrt{\tilde{g}^\top M^{-1} \tilde{g}}. \quad (\text{S46})$$

At a double stationary point, $\tilde{g} \neq 0$ is exactly the condition that some infinitesimal fixed-budget redistribution is transverse to the fold. For two controls, projected versions of the f_t and f_{tt} gradients must be linearly independent before a budget-constrained cusp unfolding is possible. These are local interior-rate statements. Positivity constraints, zero baseline rates, binary masks, and moving sharp supports require a constrained finite-amplitude design problem.

An eleven-state falsification fixture compares the basis Fréchet gradient with Eq. (S44), 256 independent budget-zero five-point derivatives, a state permutation, and 20,000 random feasible directions. The relative Duhamel error is 3.3×10^{-15} , the maximum independent finite-difference error is 2.4×10^{-8} , and the best random response is only 0.933 of Eq. (S46). At the primary finite encounter fold, the chain-rule reconstruction of the already computed log-rate transversality agrees to 6.8×10^{-15} relative error. The orthogonal near-to-far fixed-state-sum redistribution has nonzero transversality -1.1621×10^{-5} . Thus the new calculation predicts a budget-feasible unfolding direction; it does not merely refit the fold path.

S5. CHANNEL CLOCKS AND MODALITY CALCULUS

S5.1. Relative-centre GIG screening law

This screening calculation uses the special catalytic convention $\eta = D_2/D_r$, so that $C_\eta = R$ and the free relative and centre noises decouple. It is not an exact factorization of a midpoint catalyst when $D_1 \neq D_2$.

For constant drifts in free space,

$$dr = u dt + \sqrt{2D_r} dW_r, \quad dR = v_c dt + \sqrt{2D_c} dW_c. \quad (\text{S47})$$

In a one-dimensional closing geometry, let $\ell > 0$ be the initial gap and $u > 0$ the closing speed. The first-contact density of the relative process is

$$h_r(t) = \frac{\ell}{\sqrt{4\pi D_r t^3}} \exp\left[-\frac{(\ell - ut)^2}{4D_r t}\right]. \quad (\text{S48})$$

For a narrow centre patch at $z \in \mathbb{R}^d$, the free centre density is

$$p_c(z, t) = \frac{1}{(4\pi D_c t)^{d/2}} \exp\left[-\frac{|z - R_0 - v_c t|^2}{4D_c t}\right]. \quad (\text{S49})$$

Multiplying the two leading factors produces the generalized inverse Gaussian first-hitting-time family, whose use as a diffusion hitting-time law is longstanding [6]:

$$g(t; A, B, \nu) = Z^{-1} t^{-\nu} \exp(-A/t - Bt), \quad (\text{S50})$$

with, for a narrow centre patch,

$$\nu = \frac{d+3}{2}, \quad A = \frac{\ell^2}{4D_r} + \frac{|z - R_0|^2}{4D_c}, \quad B = \frac{u^2}{4D_r} + \frac{|v_c|^2}{4D_c}. \quad (\text{S51})$$

Here the initial relative displacement is $-\ell\hat{e}$ and $u > 0$ denotes closing drift along $\hat{e}$. Writing $\delta = z - R_0$, direct expansion also produces the time-independent factor $\exp[\ell u/(2D_r) + \delta \cdot v_c/(2D_c)]$. It cancels from each normalized conditional shape g_j , but it is channel- and direction-dependent and therefore contributes to the physical splitting amplitude; it cannot in general be discarded from mixture weights. The power $\nu = (d+3)/2$ is a narrow-patch screening value, not a universal exponent. Integrating over tangential or centre coordinates changes the power; reflections generate a sum of image contributions.

For $A, B > 0$,

$$Z = 2 \left(\frac{A}{B}\right)^{(1-\nu)/2} K_{1-\nu}(2\sqrt{AB}), \quad (\text{S52})$$

and the mode is

$$t_m = \frac{-\nu + \sqrt{\nu^2 + 4AB}}{2B} = \frac{2A}{\nu + \sqrt{\nu^2 + 4AB}}. \quad (\text{S53})$$

At $B = 0$, the normalizer is $Z = A^{1-\nu}\Gamma(\nu - 1)$ only for $\nu > 1$. The positive stationary mode remains $t_m = A/\nu$

for $\nu > 0$, but it defines the mode of a normalized density only when the normalizer is finite. These formulas are exact for the GIG family. Their use for a confined encounter channel is a screening approximation until a mode-window remainder is controlled.

S5.2. Fixed-shape and physical folds

Let two normalized conditional densities be g_1 and g_2 , and consider

$$f(t; w) = w g_1(t) + (1 - w) g_2(t), \quad 0 < w < 1. \quad (\text{S54})$$

A fold of the critical points satisfies $f_t = f_{tt} = 0$. Eliminating the weight gives

$$g'_1 g''_2 - g'_2 g''_1 = 0, \quad (\text{S55})$$

$$w_* = -\frac{g'_2}{g'_1 - g'_2}. \quad (\text{S56})$$

Admissibility additionally requires $g'_1 \neq g'_2$ and $0 < w_* < 1$. For $g_i = C_i t^{-\nu_i} \exp(-A_i/t - B_i t)$, define

$$\begin{aligned} a_i &= (\log g_i)' = \frac{A_i}{t^2} - B_i - \frac{\nu_i}{t}, \\ b_i &= (\log g_i)'' = -\frac{2A_i}{t^3} + \frac{\nu_i}{t^2}. \end{aligned} \quad (\text{S57})$$

Equation (S55) then becomes

$$a_1(a_2^2 + b_2) - a_2(a_1^2 + b_1) = 0. \quad (\text{S58})$$

This is exact for fixed channel shapes. A physical parameter generally changes both the weights and shapes. Its fold must be solved in the full model,

$$f_t(t_*, \theta_*) = f_{tt}(t_*, \theta_*) = 0, \quad (\text{S59})$$

with

$$a_* = f_{t\theta}(t_*, \theta_*) \neq 0, \quad b_* = f_{ttt}(t_*, \theta_*) \neq 0. \quad (\text{S60})$$

Taylor expansion yields

$$f_t = a_* \Delta\theta + \frac{b_*}{2} \Delta t^2 + o(|\Delta\theta| + |\Delta t|^2). \quad (\text{S61})$$

On the side for which $-2a_* \Delta\theta/b_* > 0$,

$$\Delta t_{\pm} = \pm \sqrt{-\frac{2a_*}{b_*} \Delta\theta} + o(|\Delta\theta|^{1/2}), \quad (\text{S62})$$

so the maximum-minimum separation scales as $|\Delta\theta|^{1/2}$, while the prominence obtained by integrating the quadratic derivative scales as $|\Delta\theta|^{3/2}$.

The regularity needed to transfer this fold between discretizations is one derivative stronger than that needed to preserve already simple modes. Let $H = (f_t, f_{tt})$. At a nondegenerate fold,

$$\det D_{(t, \theta)} H = -f_{t\theta} f_{ttt} \neq 0. \quad (\text{S63})$$

A sufficient condition for transfer is joint local C^3 convergence of $f_h(t, \theta)$. More minimally, the implicit-function argument requires C^1 convergence of $H_h = (\partial_t f_h, \partial_{tt} f_h)$, including the specific time and parameter derivatives entering DH_h . These two regularity statements are not equivalent. Under either sufficient set of hypotheses, the implicit-function theorem transfers a unique nearby non-degenerate fold. Mere C^2 convergence of the densities is not sufficient. Indeed,

$$f(t, \theta) = \theta t + t^3/3, \quad f_n = f + 2n^{-3} \sin(nt) \quad (\text{S64})$$

satisfies $f_n \rightarrow f$ in C^2 , but the nearby double stationary point $(t_n, \theta_n) = (0, -2/n^2)$ has $f_{n,ttt} = 0$ and is degenerate. The continuum bridge therefore requires uniform semigroup and parameter derivatives through third time order near the fold, not only convergence of the density curves.

For three channels, a positive simplex solution of

$$\begin{pmatrix} 1 & 1 & 1 \\ g'_1 & g'_2 & g'_3 \\ g''_1 & g''_2 & g''_3 \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad (\text{S65})$$

locates only a candidate double stationary point. It is a nondegenerate fixed-shape fold only if $\sum_i w_i g'''_i(t_*) \neq 0$ and a declared one-parameter control direction also has $f_{t\theta}(t_*) \neq 0$. Equation (S65) alone supplies neither condition and does not prove trimodality. Existence of three resolved modes requires five alternating simple critical points with positive margins and tail control. Only a claim of an exact global root or mode count additionally requires interval-exhaustive root isolation. No cusp or trimodal claim follows from Eq. (S65) alone; the independent finite-state certificate in Sec. S7.6 supplies the existence checks for one declared bounded family while explicitly withholding a global root-count theorem.

S6. FINITE LATTICE AND CTMC MECHANISM CERTIFICATE

S6.1. Synchronous discovery family

For independent one-step kernels P_1 and P_2 , let $P = P_1 \otimes P_2$. With U selecting catalytic co-locations and $D_\rho = \text{diag}(\rho_1, \rho_2)$, the arrival-then-react convention gives

$$Q = P(I - UD_\rho U^\top), \quad B = PUD_\rho, \quad (\text{S66})$$

$$f_j(n+1) = \alpha Q^n B e_j. \quad (\text{S67})$$

Here α is live mass conditioned on surviving any time-zero reaction check; the convention includes no $n = 0$ reaction atom, and the first reported flux is at step one. The discovery starts away from catalytic states, so this conditioning does not alter its saved curves. The generating function is

$$\hat{f}(z) = z\alpha(I - zQ)^{-1}B\mathbf{1}. \quad (\text{S68})$$

The discovery family uses reflecting attempted steps, equal reaction probabilities $\rho_1 = \rho_2 = 0.6$, movement probabilities $(q_1, q_2) = (0.7, 0.2)$, and right biases 0.3. The near catalyst and starting configuration are translated with system size while a far catalyst remains at the right boundary. Figure S1 shows a stable early chase channel and a late boundary-transport channel. Across $L = 31, 41, 61, 81$, the early peak remains at $n = 33$, while the late peak moves from 194 to 598. The near-channel weight approaches 0.05369, the valley ratio falls from 0.308 to 9.20×10^{-4} , and all 155 multiscale morphology views retain two peaks. Tail mass is below 10^{-12} . A local perturbation box gives 37 bimodal cases among 40 draws; the remaining three are shoulder or unimodal cases and are not discarded.

Poissonization and an independent-clock CTMC retain the two clock windows (Fig. S2), but they are robustness tests rather than a calibrated continuous-time limit of the synchronous chain. In particular, rate and time parameters must not be exchanged between these models.

S6.2. Numerically located finite-CTMC fold

In the independent-clock $L = 31$ model, we vary

$$\theta = \log(\kappa_{\text{near}}/\kappa_{\text{far}}), \quad (\text{S69})$$

holding $\kappa_{\text{far}} = 0.9162907319$ fixed. Because θ changes both T and the reaction observable, this is a physical parameter path, not a freely adjusted mixture weight. The finite-matrix time-derivative identities follow Eq. (S34). Parameter derivatives are obtained by propagating the column state and sensitivity,

$$p' = T^\top p, \quad s' = T^\top s + T_\theta^\top p, \quad s(0) = 0, \quad (\text{S70})$$

in one augmented exponential.

The fold occurs at

$$t_* = 37.0749586402, \quad \theta_* = -9.6753635853, \quad (\text{S71})$$

or $\kappa_{\text{near}} = 5.755410148 \times 10^{-5}$. Dimensionless first- and second-derivative residuals are 7.1×10^{-15} and 6.8×10^{-13} ; the dimensionless third derivative is 69.94 and parameter transversality is -0.1962 . Independent points, which were not used to locate the fold, give log-log slopes 0.50095 for critical-point separation and 1.50877 for prominence (Fig. 2 of the main article). The mass-closure error is 4.0×10^{-15} , and survival at $t = 5000$ is 1.3×10^{-21} .

For the canonical channel comparison, the joint relative-centre GIG predicts the early mode at 26.50, versus 32.1534 for the CTMC, and the inverse-Gaussian boundary clock predicts 180.19, versus 196.1459. The CTMC values are Brent roots of the analytical channel derivative $\alpha e^{Tt} T b_j$, with negative second derivatives, rather than maxima of a sampled time series. The relative errors are 17.6% and 8.1%, respectively. These

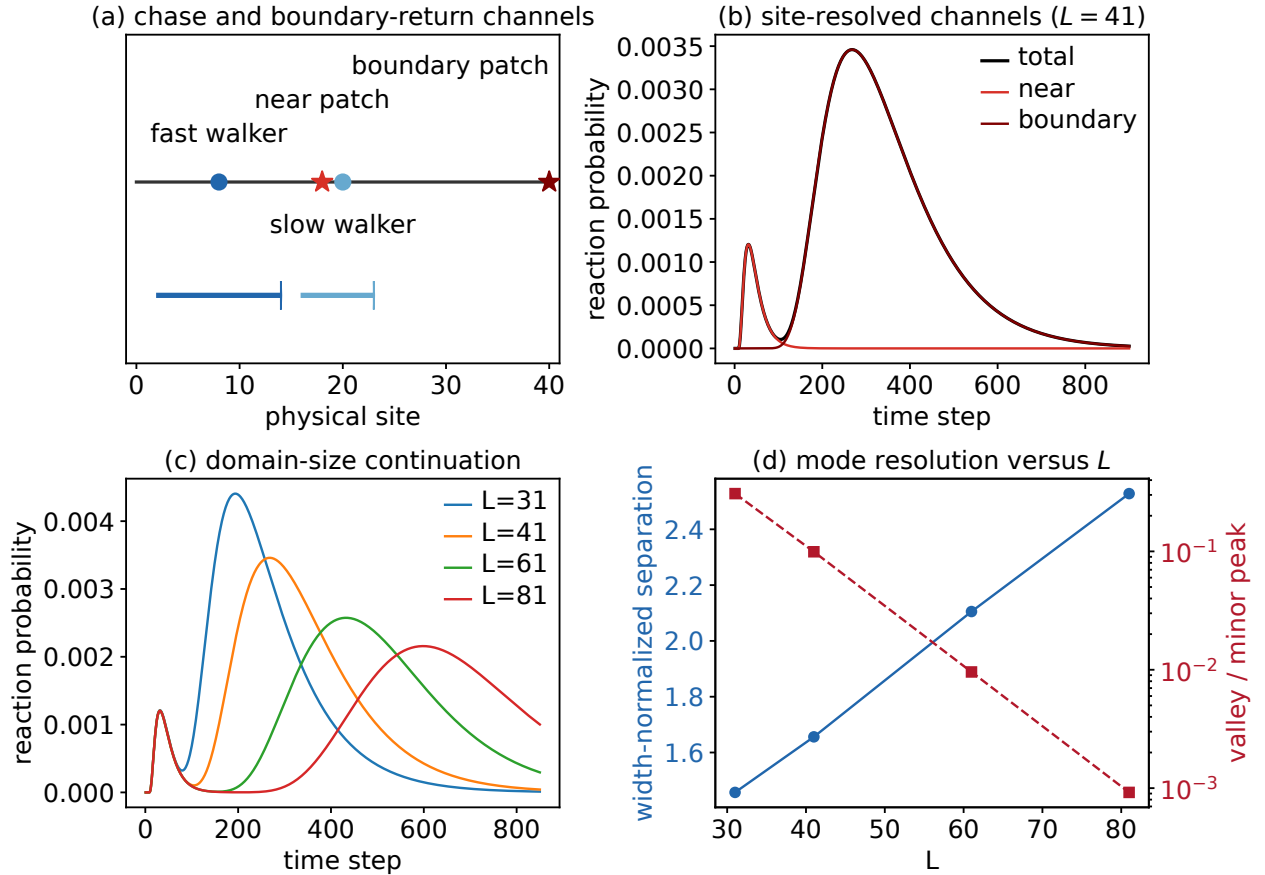


FIG. S1. Finite synchronous discovery family. The total reaction-time mass is resolved into a weak early chase channel at the near catalytic site and a strong late channel associated with transport to the far site. The early peak and splitting weight stabilize while the late peak moves with domain length. The displayed finite-size trends are descriptive evidence, not an asymptotic theorem.

non-negligible errors are reported because no uniform reflected-path remainder is available. The screening law organizes parameter search; it does not replace the direct finite-matrix semigroup calculation.

S7. FINITE-RADIUS TWO-DIMENSIONAL REALIZATION

S7.1. Discretization and morphology audit

The physical domain is the obstacle-free unit square with reflecting outer boundaries. Each one-particle generator is a conservative boundary-node lattice CTMC with grid spacing $L/(n-1)$, central nearest-neighbour diffusion, upwind longitudinal drift, a smooth transverse restoring drift, and reflection by omission of outward jumps. This is not a cell-centred finite-volume scheme: physical D, v, γ, κ have dimensions $L^2/T, L/T, T^{-1}, T^{-1}$, respectively. Length and time are scaled by the square side L and a declared time T , and the reported numbers are the dimensionless groups $DT/L^2, vT/L, \gamma T, \kappa T$. The

geometric state-sum quadrature assigns equal weights to nodes, and contact and patch supports are binary node tests. With drift, that equal-node measure is not the CTMC stationary distribution. Accordingly, the matched catalyst budget below is an unweighted state-wise sum, not a stationary-exposure match. The free pair generator is the Kronecker sum of the two one-particle generators. Killing is applied whenever the pair distance is below the physical reaction radius $a = 0.13$ and the declared midpoint $C_{1/2} = (X_1 + X_2)/2$ lies inside a catalytic patch. In the continuum, replacing the open contact and patch disks by their closures changes only zero-measure boundaries. On every saved lattice, however, both contact and patch masks use closed “less-than-or-equal” tests with a declared scale-aware floating-point length tolerance. This convention can change binary state counts at a grid boundary, so those counts and the same closed-mask convention are retained in the machine artifacts and contact-safe diagnostics. Bilinear initial weights preserve physical starting positions across grids, but their independent product can put artificial mass in the discrete contact tube on a coarse lattice even

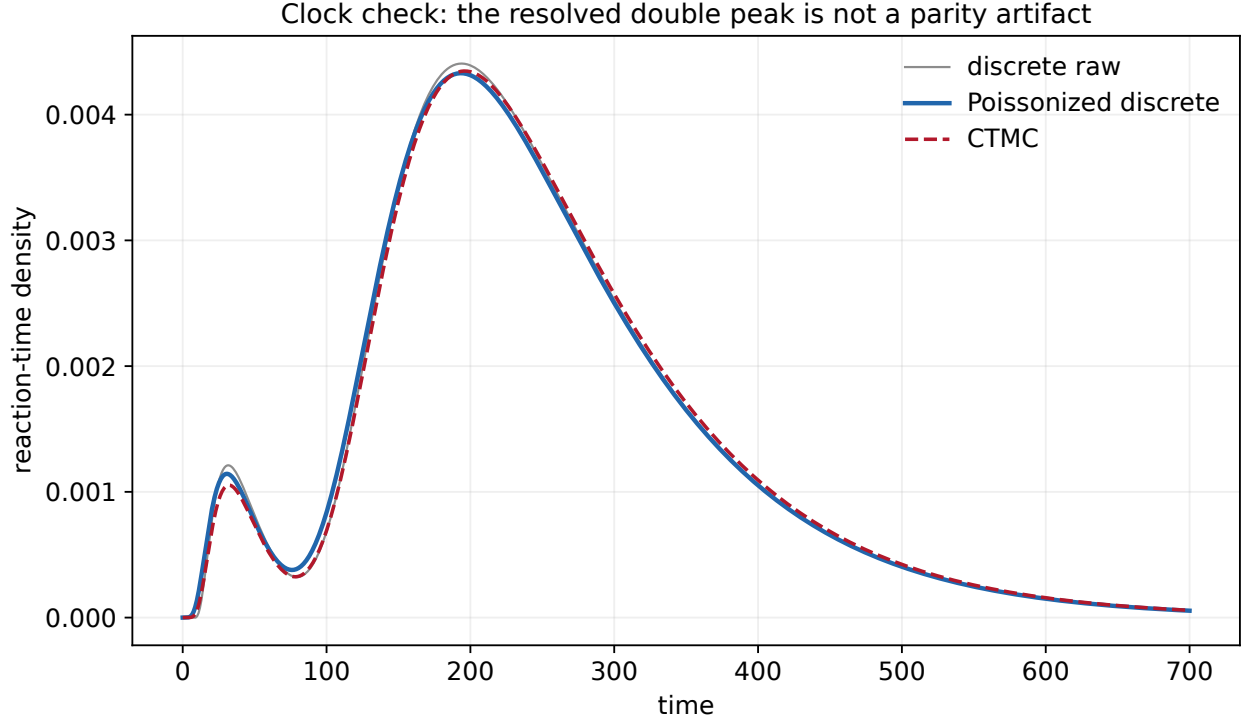


FIG. S2. Clock robustness. Poissonized synchronous dynamics and an independent-clock CTMC retain separated early/late windows. The curves test mechanism robustness under clock changes; they do not establish a calibrated synchronous-to-CTMC continuum limit.

when the two physical starts satisfy $|x_1^0 - x_2^0| > a$. We therefore use the product weights unchanged when they are contact-safe and otherwise construct a joint initial law by the nonnegative programme

$$\begin{aligned}
 & \underset{\alpha_{ij}}{\text{minimize}} && \sum_{ij} \alpha_{ij} (|x_i - x_1^0|^2 + |x_j - x_2^0|^2), \\
 & \text{subject to} && \sum_{ij} \alpha_{ij} = 1, \\
 & && \sum_{ij} \alpha_{ij} x_i = x_1^0, \\
 & && \sum_{ij} \alpha_{ij} x_j = x_2^0, \\
 & && \alpha_{ij} \geq 0, \\
 & && \alpha_{ij} = 0 \quad \text{if } |x_i - x_j| \leq a + \tau_\ell.
 \end{aligned} \tag{S72}$$

Here $\tau_\ell = 64\epsilon_{\text{mach}} \max(L_x, L_y, a)$ is the binary64 scale-aware length guard used by the saved contact mask. Patch disks use the analogous guard with their radius and centre coordinates included in the maximum. Thus Eq. (S72) and the implementation use the same tolerant closed-support convention. The smallest feasible expanding local stencil is selected. Because the minimum-spread linear programme can have a degenerate optimal face, a second, strictly convex programme selects on that face the unique law closest in Euclidean norm to the original

bilinear product weights; the result therefore does not depend on an arbitrary linear-programming basis. Physical starts in the closed contact region are rejected. Saved diagnostics verify unit mass, nonnegativity, both vector means, zero initial mass on the contact and active-sink supports, and the root-mean-square and maximum displacement of each represented start. The coarse-grid solution need not retain independent marginals; it is the declared minimum-spread, closest-product representation of two deterministic, noncontacting starts. Consequently, all results in this section are finite-state mechanism certificates; no continuum claim is deduced from the boundary discretization.

The principal matched-control family uses

$$(D_1, v_1) = (0.0025, 0.18), \quad (D_2, v_2) = (0.0008, 0.02), \tag{S73}$$

transverse confinement 1.5, starts (0.10, 0.50) and (0.35, 0.50), and patterned reactivity

$$\kappa_{\text{pat}}(C_{1/2}) = 0.5 \mathbf{1}_{C_{\text{near}}}(C_{1/2}) + 15 \mathbf{1}_{C_{\text{far}}}(C_{1/2}), \tag{S74}$$

with C_{near} centered at (0.25, 0.50), radius 0.18, and C_{far} centered at (0.72, 0.50), radius 0.20. The far patch has positive boundary clearance 0.08.

To prevent controls from being mistaken for one-factor ablations, Table SII assigns each numerical family an immutable identifier. All four use $(D_1, D_2) =$

(0.0025, 0.0008), transverse confinement 1.5, midpoint catalysis, and patch y -coordinate 0.5; the entries $x/r/\kappa$ specify patch centre, radius, and rate. The families are separate examples unless an explicit matched path is stated.

At each grid, a homogeneous rate $\bar{\kappa}_h$ is chosen so that

$$\sum_x K_{\text{hom}}(x) = \sum_x K_{\text{pat}}(x) \quad (\text{S75})$$

on the same discrete encounter tube. This is an equal per-lattice discrete state-sum killing budget. It is not equality of every trajectory-level hazard, splitting probability, or mean lifetime.

For the midpoint coordinate, the change of variables $C = (x_1 + x_2)/2$, $r = x_1 - x_2$ has unit Jacobian. In the unit square, for $0 \leq a \leq 1$, the exact finite-radius volume of the contact tube is

$$V_T(a) = \int_{|r| \leq a} (1 - |r_x|)(1 - |r_y|) dr = \pi a^2 - \frac{8}{3}a^3 + \frac{1}{2}a^4. \quad (\text{S76})$$

If every catalyst disk has boundary clearance at least $a/2$, its centre-space section is untruncated for every admissible r . The finite- a continuum counterpart of Eq. (S75) is then

$$\bar{\kappa}_{h,\text{cont}}(a) = \frac{\pi a^2 [\pi \sum_j \kappa_j \rho_j^2]}{V_T(a)}. \quad (\text{S77})$$

For M2D-E, $a = 0.13$, both clearances exceed $a/2$, and $\bar{\kappa}_{h,\text{cont}} = 2.169402271$. The present 17×13 state-sum rate is 1.849069451, still 14.8% below this reference; the line in Fig. S3 is therefore a finite-radius quadrature target, not a claim that the five dynamics grids have reached the continuum.

The density is propagated by sparse matrix exponentials on $0 \leq t \leq 80$ with spacing 0.1. The audit uses raw and binned densities, smoothing windows 1, 3, 5, 9, 15, bin widths 1, 2, 4, 8, 16, prominence, lobe mass, valley depth, and persistence across 155 scale views. Survival is propagated separately to $t = 960$, and the post-window density is searched for additional extrema. The discrete generator satisfies channel mass balance to 3.8×10^{-15} or better. Trapezoidal time integration is used only as a secondary closure diagnostic.

S7.2. M2D-E equal-budget patterned and homogeneous endpoints

Figure S3 compares the M2D-E patterned model with its homogeneous equal-budget counterpart. The four finer patterned grids are classified resolved-bimodal, with early peaks 1.3–1.5 and late peaks 19.2–20.8. The 9×5 patterned grid is a resolved shoulder: its detected roots, refined by direct finite-matrix semigroup derivative evaluations, give strict maxima at 0.814 and 17.816, but only one classifier mode. All homogeneous counterparts are

classified resolved-unimodal, with an accepted peak 1.1–1.2. An independent finite-matrix semigroup derivative audit retains a strict late secondary maximum in each homogeneous model, but its height is only 1.02%–1.81% of the primary peak; the patterned secondary/primary ratios are 4.43%–8.15%. A persisted 101-point sweep of the full classifier at cutoffs strictly inside the unrounded interval (1.809905%, 4.434632%) retains every displayed label: the homogeneous cases remain unimodal, the four finer patterned cases remain bimodal, and the coarse patterned case remains a shoulder. The interval endpoints were computed from these same patterned and control realizations, so this is a post-hoc finite-sample separation audit. It is neither a predeclared detector range nor an experiment-specific noise-robustness result, and it is not a continuous-interval theorem. Repeating the homogeneous control with tensor-product boundary control-volume budget weights gives strict secondary ratios 0.637%–1.217%, zero two-peak classifier views on every grid, and the same non-bimodal resolution class (one morphology is a shoulder). Thus the resolution-class contrast is not an artifact of the state-sum budget measure. Spatial patterning promotes a subthreshold transport clock to a resolved mode; it is not claimed to create the mathematical existence of a second local maximum. The discrete budget mismatch is at most 4.2×10^{-16} . No post-window maxima are found, and the propagated tail at $t = 960$ is at most 1.23×10^{-10} for the patterned family and 8.54×10^{-30} for the homogeneous family. Table SIII records the gridwise peak calls and verifies the state-sum budget equality separately on every lattice; the figure also shows the control-volume-matched rates.

These are finite-lattice comparisons, not a continuum convergence study. For M2D-E walker 1, the longitudinal cell Péclet number decreases only from 9.0 to 4.5, so the upwind effective-diffusion ratio $D_{\text{eff}}/D = 1 + \text{Pe}_h/2$ decreases from 5.5 to 3.25. Even one transverse cell from the centre on the finest lattice gives local Péclet numbers 4.17 and 13.02 for walkers 1 and 2. These diagnostics block an SDE-refinement interpretation while leaving the declared finite CTMC comparison exact.

S7.3. M2D-F matched-budget fold continuation

For the separate M2D-F family, a natural physical continuation redistributes its fixed per-lattice state-sum budget,

$$\kappa_\theta(C_{1/2}) = (1 - \theta)\bar{\kappa}_h + \theta\kappa_{\text{pat}}(C_{1/2}), \quad 0 \leq \theta \leq 1. \quad (\text{S78})$$

Along this path, both splitting weights and conditional shapes vary. To resolve the interior fold directly, we use a second declared family with the same patch geometry and rates but reaction radius $a = 0.17$, walker parameters $(D_1, v_1) = (0.0025, 0.115)$ and $(D_2, v_2) = (0.0008, 0.02)$, and starts $(0, 0.5)$ and $(0.28, 0.5)$. The distinction from the endpoint-ablation family above is explicit: the fold

TABLE SII. Declared bounded two-dimensional model families and M2D-C branches. Starts list the two initial x -coordinates; all initial y -coordinates are 0.5. The uniform M2D-C row is the unit-domain member of the separately audited domain-length sweep.

ID	Role	a ; starts	(v_1, v_2)	Patches $x/r/\kappa$
M2D-E	matched endpoints	0.13; 0.10, 0.35	0.18, 0.02	0.25/0.18/0.5; 0.72/0.20/15
M2D-F	matched fold	0.17; 0, 0.28	0.115, 0.02	0.25/0.18/0.5; 0.72/0.20/15
M2D-C	separated boundary	0.13; 0.10, 0.35	0.18, 0.02	0.28/0.12/0.2; 0.90/0.20/4
M2D-C	interior two-patch	0.13; 0.10, 0.35	0.18, 0.02	0.28/0.12/0.2; 0.75/0.18/4
M2D-C	single far	0.13; 0.10, 0.35	0.18, 0.02	0.90/0.20/4
M2D-C	co-located labels	0.13; 0.10, 0.35	0.18, 0.02	0.90/0.20/0.2; 0.90/0.20/4
M2D-C	uniform (unit domain)	0.13; 0.10, 0.35	0.18, 0.02	0.50/2.00/0.2
M2D-T	three-patch trimodality	0.13; 0.05, 0.20	0.10, 0.02	0.20/0.06/0.03; 0.70/0.05/1; 0.94/0.05/0.05

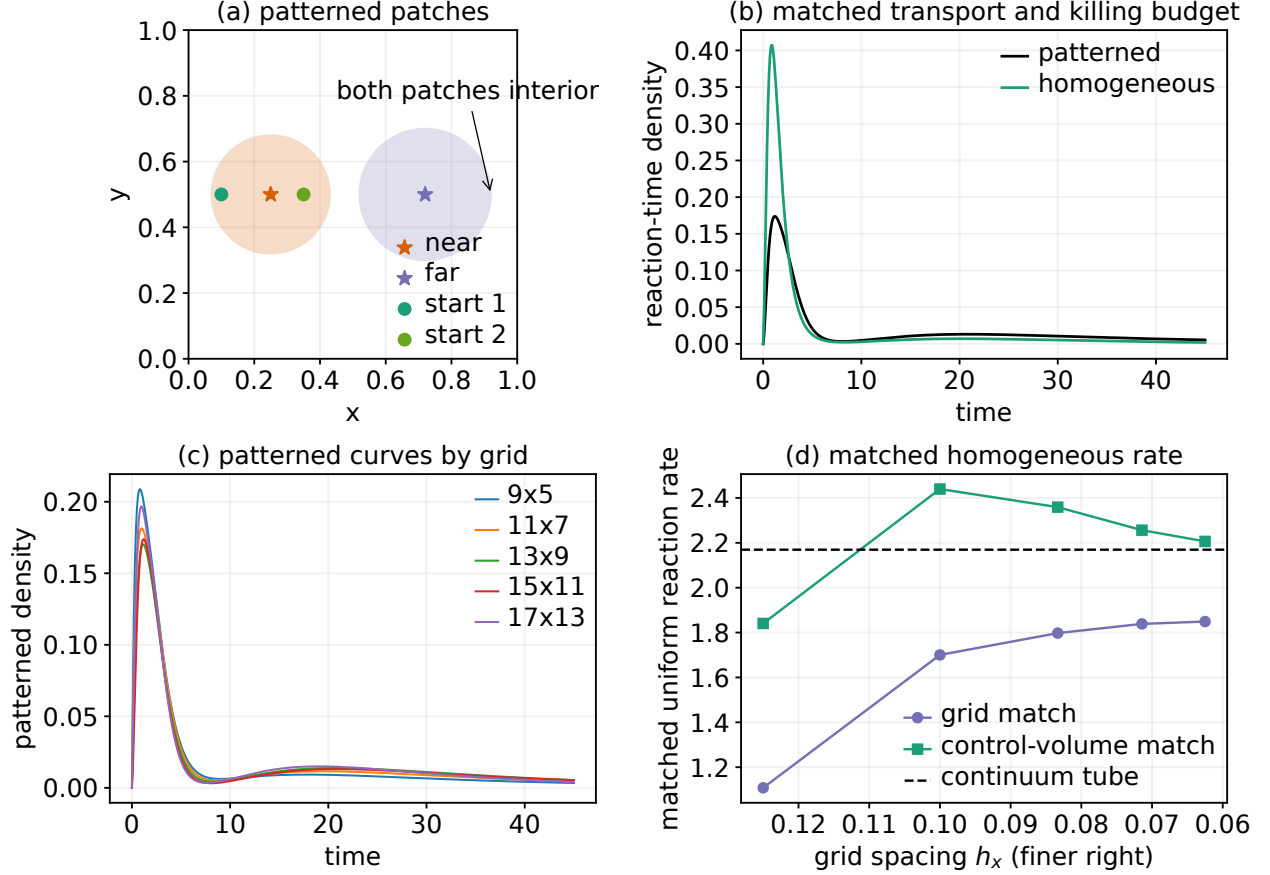


FIG. S3. M2D-E finite-radius two-dimensional matched control. At each grid, the patterned and homogeneous fields have the same sum of statewise killing rates on the same encounter tube. Homogeneous endpoints are resolved-unimodal on all five grids; patterned endpoints are resolved-bimodal on the four finer grids and a resolved shoulder on the coarsest. All strict stationary points, including the coarse patterned and small homogeneous secondary maxima, are retained separately. A separate product-control-volume match retains the non-bimodal homogeneous controls. This establishes finite-grid amplification of a transport clock under spatial redistribution of a fixed discrete killing budget; it does not, by itself, locate or prove a continuum fold.

family is a mechanism certificate, not a post hoc claim that every matched endpoint path has the same critical coordinate. In particular, M2D-F is not a continuation of M2D-E: its transport parameters, reaction radius, starts,

and per-lattice budget path differ, so neither a common continuation nor a shared critical coordinate is claimed. On each grid, $\bar{K}_h$ is recomputed from the reactive-state counts, so $\sum_x K_\theta(x)$ is constant along θ to roundoff.

TABLE III. M2D-E two-dimensional equal-budget endpoint audit. N_+ is the number of multiscale views containing two accepted peaks out of 155. The displayed homogeneous column uses state-sum matching; the control-volume-matched homogeneous model also has $N_+ = 0$ on every grid. “Strict maxima” lists both detected, derivative-refined patterned maxima even when the classifier calls a shoulder. Each homogeneous model also has a small strict secondary maximum in the machine artifact; the displayed homogeneous column lists only the classifier-accepted peak.

Grid	Patterned call	Patterned strict maxima	Homogeneous accepted peak	N_+ patterned/homogeneous	Relative budget error
9×5	shoulder	0.814, 17.816	1.1	123/0	4.10×10^{-16}
11×7	bimodal	0.990, 19.518	1.1	153/0	3.08×10^{-16}
13×9	bimodal	1.142, 20.679	1.1	155/0	3.51×10^{-16}
15×11	bimodal	1.191, 20.831	1.2	155/0	0
17×13	bimodal	0.962, 19.061	1.1	152/0	2.23×10^{-16}

Sparse generator actions give f_t, f_{tt}, f_{ttt} , while an augmented matrix exponential gives the parameter derivatives. On the 9×5 grid the fold is

$$(t_c, \theta_c) = (16.8093321, 0.27537300), \quad (\text{S79})$$

with residuals below 3.5×10^{-17} , $f_{ttt} = -1.34 \times 10^{-5}$, and $f_{t\theta} = 1.67 \times 10^{-4}$. On the 11×7 grid the fold is

$$(t_c, \theta_c) = (18.0995323, 0.01381035), \quad (\text{S80})$$

with residuals $(|f_t|, |f_{tt}|) = (8.7 \times 10^{-19}, 3.5 \times 10^{-18})$, $f_{ttt} = -7.81 \times 10^{-6}$, and $f_{t\theta} = 2.83 \times 10^{-4}$. On the 13×9 grid,

$$(t_c, \theta_c) = (16.5807587, 0.25589201), \quad (\text{S81})$$

with residuals below 2.8×10^{-17} , $f_{ttt} = -5.96 \times 10^{-6}$, and $f_{t\theta} = 7.46 \times 10^{-5}$. Independent continuation gives separation and prominence exponents $(0.4989, 1.5001)$, $(0.4933, 1.4843)$, and $(0.4993, 1.5039)$, respectively. Thus all three finite generators verify the nondegenerate normal form. They are three separate finite-model mechanism certificates, not a convergent refinement sequence or an estimator of a continuum critical control.

The state-count-matched critical controls span 0.262 and are nonmonotone. As a one-factor budget-measure audit, we replace equal node weights by the tensor-product boundary trapezoidal weights on the four-dimensional two-walker domain, without changing the CTMC generator or reaction masks. The three nondegenerate folds persist at $\theta_c = 0.6244, 0.2408, 0.4593$, but remain nonmonotone and span 0.384. The M2D-F clearances 0.070 and 0.080 are slightly below $a/2 = 0.085$, so Eq. (S77) is not exact for this family. Direct one-dimensional quadrature of the boundary-clipped midpoint sections gives $\bar{\kappa}_{h,\text{cont}} = 2.2501572$; the factorized interior-patch counterfactual is 2.2502051, a relative difference of only 2.13×10^{-5} . The weighted homogeneous rates lie closer to the clipped reference, yet their fold sensitivity confirms that neither sequence estimates a continuum critical value.

The two even grids clarify the observed topology without supplying a root-count theorem. On 12×8 , a

nondegenerate continuation root occurs at $(t_c, \theta_c) = (18.26877, -0.06158)$, outside the physical interval. On 10×6 , a declared scan of the $f_{tt} = 0$ branches over $t \in [0.01, 40]$, $\theta \in [-0.2, 0.1]$, followed by four bounded least-squares starts, finds no fold; every optimizer reaches the same positive near miss with $f_t = 3.8721 \times 10^{-6}$. This is only “no root found under the declared search,” not proof of absence.

A separate endpoint audit on $9 \times 5, 10 \times 6, 11 \times 7, 12 \times 8, 13 \times 9$ grids retains every detected sign-changing stationary point and calls a secondary maximum resolved only if its height exceeds 3% of the primary peak and its intervening valley is below 95% of the smaller maximum. The finite scan does not certify the absence of tangential or narrower roots between sampled times. All homogeneous endpoints are resolved-unimodal and all patterned endpoints are resolved-bimodal. The two even grids contain homogeneous reflection ripples below the declared resolution threshold; these roots are reported rather than deleted. All ten endpoint tails at $t = 480$ are below 1.9×10^{-6} . The fixed-budget path, endpoint audit, finite-grid tangencies, and independent normal-form scaling are collected in Fig. 3 of the main article and are not duplicated here.

S7.4. Catalytic-coordinate sensitivity

Equation (S11) shows that midpoint catalysis and the diffusion-decoupling coordinate define different sinks when $D_1 \neq D_2$. We therefore repeat the M2D-E patterned endpoint calculation with only

$$\eta : \frac{1}{2} \longrightarrow \frac{D_2}{D_1 + D_2} = 0.242424 \dots \quad (\text{S82})$$

changed. Since this alters the discrete patch support and its raw state-sum budget, the homogeneous control is rematched independently within every (grid, η) pair. All six homogeneous controls remain resolved-unimodal under the same canonical threshold; small strict secondary maxima are retained by the derivative audit. The

midpoint and weighted patterned labels are respectively shoulder/shoulder on 9×5 , bimodal/bimodal on 11×7 , and bimodal/bimodal on 13×9 . Near-patch mask Jacard distances are 0.2222, 0.1000, and 0.2295; far-patch distances are 0.1000, 0.0769, and 0.1493. Thus the catalytic coordinate is an observable model convention, not a relabeling. Label agreement is not model equivalence: the masks and densities differ on every grid. The finest two comparisons support persistence of the bimodality mechanism under this diagnostic change, but the coarse shoulder, nonmonotone masks, and short sequence preclude a continuum-robustness claim.

A supplemental 13×9 M2D-T calculation remains three-mode for both coordinates even though its three patch-mask distances are 0.556, 0.333, and 0.500. This check is reported separately and does not merge the two physical models. Figure S4 displays the one-factor mask and morphology sensitivity audit.

S7.5. Interior and adverse controls

The separately declared M2D-C examples in Fig. S5 illustrate spatial reaction selection and transport clocks. They change the patch geometry and rates listed in Table SII and therefore are not factorial one-parameter ablations of M2D-E or M2D-F. Removing the near patch within M2D-C gives a resolved-unimodal single-far case. Co-locating the two channel labels at the far position is also resolved-unimodal, showing that two labels are not sufficient for a resolved second mode. Direct finite-matrix semigroup derivative evaluations retain a shallow strict max–min–max pattern in both controls; the second peak fails the scale-persistence and valley-depth criteria (not the standalone prominence threshold). Moving the far patch to centre $x = 0.75$, radius 0.18, leaves a strictly positive clearance 0.07; the resulting family remains bimodal on the 9×5 through 15×11 grids. This is a non-touching interior control, not proof that the reflecting wall is dynamically irrelevant: late trajectories can still visit the wall.

Most importantly, a separate spatially uniform reactivity $\kappa = 0.20$ is bimodal on the 11×7 grid, with accepted peaks near 1.3 and 24.0. Domain-length scans keep the early peak near one and move the late peak linearly; the fitted late-peak slope times the slow drift is 0.983, which is consistent with a first-pass/boundary-return interpretation. This scaling fit is descriptive over the finite $0 \leq t \leq 160$ window: the $L_x = 3$ survival is still 0.152 at its endpoint. A separate post-window audit propagates every domain-scan member to $t = 480$, where the maximum survival is below 6.8×10^{-9} . On the declared 0.25-spaced audit mesh, direct finite-matrix derivative observables satisfy $f_t < 0$ and $f_{tt} > 0$, with no detected post-window stationary point. This closes the stated tail gate but is neither an interval-exhaustive root proof nor a pathwise clock decomposition. Thus bimodality persists under uniform reactivity, and reac-

tion patterning is not necessary. The matched result in Sec. S7.2 consequently supports the restricted statement that *spatial redistribution of a fixed budget can promote a subthreshold transport clock into a resolved mode in the stated family*. It does not support a universal necessity claim or creation of a previously nonexistent strict extremum.

S7.6. Three-patch finite-grid trimodality

The separate M2D-T family tests whether three non-negative spatial channels can produce three resolved modes in a bounded finite-radius encounter model. Its parameters are listed in Table SII. Direct channel-flux dominance associates the three detected maxima, in time order, with the near, middle, and far patches. Their timing is consistent with short-launch, advective, and delayed downstream-return interpretations, respectively, but the calculation does not provide a pathwise occupancy decomposition. The far disk remains inside the square with clearance 0.01. This geometry is deliberately reported as a finite-lattice construction: the patch radii 0.06, 0.05, 0.05 are smaller than h_x on all four lattices. Their near/middle/far reactive-state counts are respectively 3/3/2, 5/5/3, 4/4/3, and 18/5/5; the nonmonotone counts expose binary-mask aliasing rather than continuum refinement.

On each of the 9×5 , 11×7 , 13×9 , and 15×11 grids, the raw density has three strict maxima. The canonical multiscale classifier resolves all three on the latter three grids; on 9×5 it retains many three-peak scale views but makes the conservative one-mode “shoulder” call. Independently, the exact finite-state identity

$$f^{(j)}(t) = \alpha e^{Tt} T^j b \quad (\text{S83})$$

is used to scan f_t to $t = 2000$ and refine every sign-change bracket. The declared scan detects five simple sign-changing roots on each grid, establishing at least five positive-time roots in the alternating order maximum–minimum–maximum–minimum–maximum (Table SIV). Here “detected” is essential: the finite sign-change scan and off-root logarithmic-slope diagnostic do not constitute an interval-certified exclusion of an additional tangential root or unresolved narrow root pair. The three resolved maxima themselves, their intervening minima, and their channel attribution do not rely on such an exhaustiveness claim. At the three maxima the dominant flux channels are, in order, near, middle, and far, with minimum dominant share 0.721. The maximum absolute root residual is 1.73×10^{-15} , the smallest raw adjacent prominence is 0.124 relative to the primary peak, and the largest remaining survival at $t = 2000$ is 4.20×10^{-11} .

Figure S6 shows the five-root alternation and the direct near/middle/far channel attribution across the four declared finite grids.

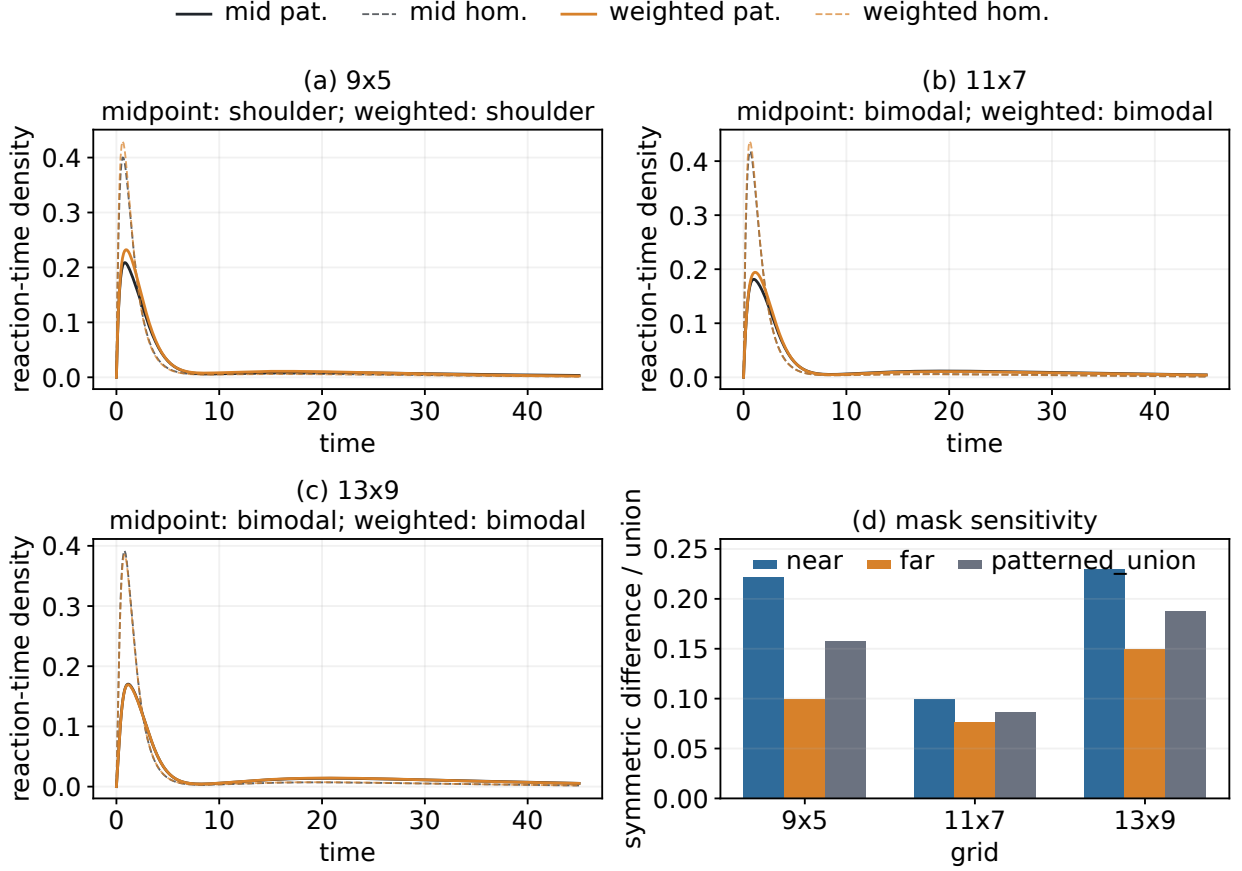


FIG. S4. Catalytic-coordinate sensitivity. The physical midpoint and the diffusion-decoupling weighted coordinate produce different finite-state patch masks and densities. Both conventions give a shoulder on the coarsest grid and bimodality on two finer grids; every within-coordinate budget-matched homogeneous control is resolved-unimodal under the declared classifier. This is a finite-grid sensitivity audit, not evidence that the two models are equivalent in a continuum limit.

TABLE SIV. The five listed positive-time detected sign-changing derivative roots for the M2D-T family establish at least five roots on each grid. Each maximum column is dominated respectively by the near, middle, and far Doi flux; the finite scan is not interval-exhaustive.

Grid	maxima	intervening minima
9×5	1.122294, 8.554300, 34.329213	3.701089, 17.850451
11×7	1.013153, 8.264158, 38.893714	3.898736, 16.798381
13×9	1.016730, 8.936256, 48.281973	4.455170, 15.934062
15×11	0.854207, 9.315094, 48.836751	4.951229, 15.557974

S8. CAPACITY-CALIBRATED DIMENSIONAL EXTENSION

Point co-location is not a mesh-independent reaction prescription in physical dimension $d \geq 2$. The encounter tube has normal dimension d , even though configuration space has dimension $2d$. That codimension selects the small-radius capacity law.

S8.1. Two-dimensional logarithmic capacity

Locally in the relative coordinate,

$$G(r, r') \sim -\frac{1}{2\pi D_r} \log |r - r'| + H \quad (d = 2), \quad (\text{S84})$$

so the effective conductance of a small reactive cross-section is inverse-logarithmic,

$$k_{\text{eff}}^{(2)}(a) \simeq \frac{2\pi D_r}{\log[\ell/(a \text{ cap}_{\log} K)] + \beta_{\text{int}} + 2\pi D_r H}. \quad (\text{S85})$$

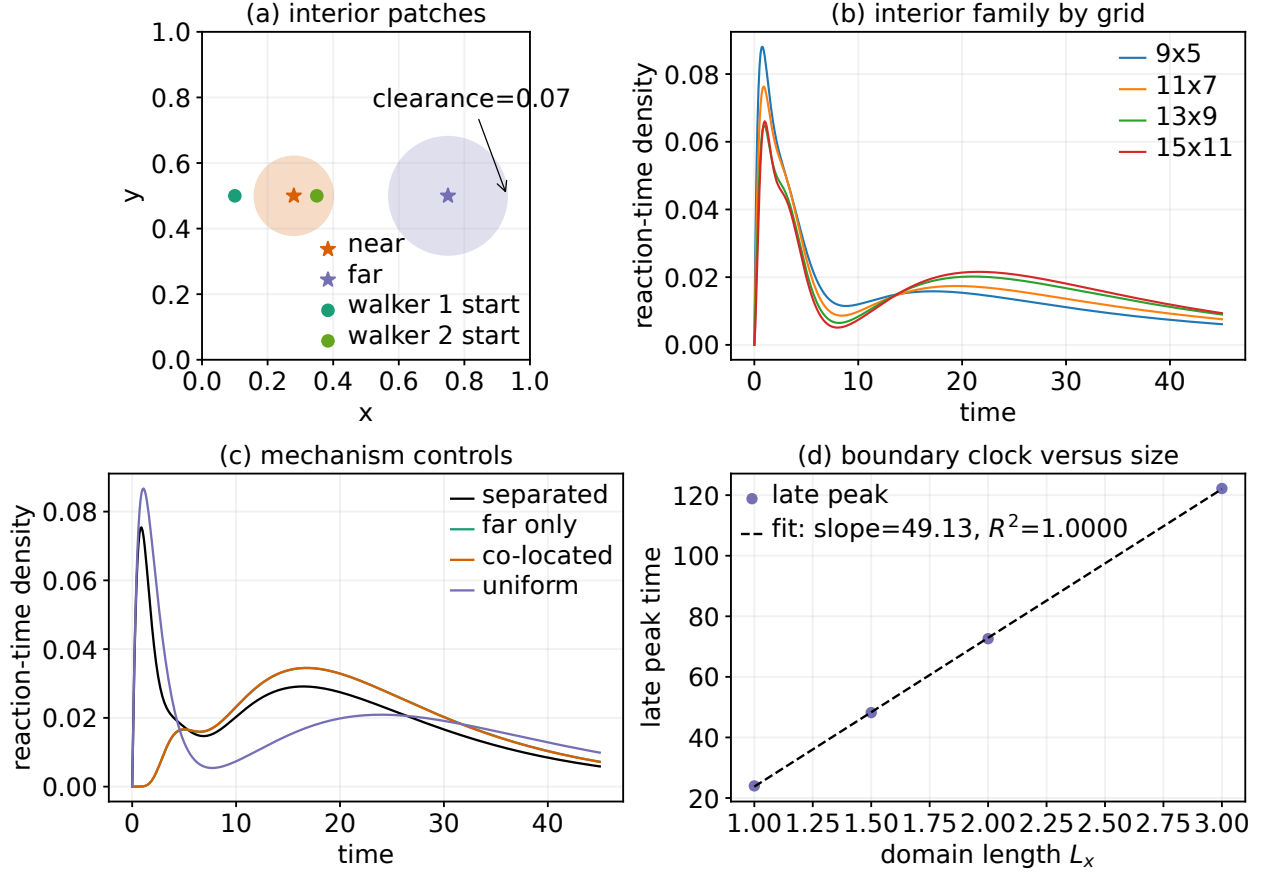


FIG. S5. Separate mechanism examples in the finite-radius two-dimensional model. The non-touching interior two-patch family remains resolved-bimodal; single and co-located reaction opportunities are resolved-unimodal despite retained shallow strict extrema. A different uniformly reactive model is bimodal, with its descriptive domain-length scaling consistent with a first-pass and boundary-return interpretation. A separate tail and finite-matrix derivative audit through $t = 480$ finds no later stationary point, but is not interval-exhaustive. These are M2D-C examples rather than one-factor ablations of the matched M2D-E family. They fix the causal language: patterning is one controllable route, not a necessary condition.

For a circular Doi disk with $\lambda = a\sqrt{\kappa/D_r}$, a local intrinsic impedance benchmark is

$$\beta_{\text{int}}^{\text{Doi}} = \frac{I_0(\lambda)}{\lambda I_1(\lambda)}. \quad (\text{S86})$$

The domain regular part H , drift, and stationary centre weight cannot be dropped in the full patterned problem.

To test the normal capacity law without centre heterogeneity, both particles are placed on the same flat two-torus with translation-invariant reaction. The pair problem then reduces exactly to a two-dimensional relative process with diffusivity $D_r = D_1 + D_2$. At fixed $\chi = \kappa a^2/D_r = 1$, the unit-area mean time has leading slope

$$\langle T \rangle = \frac{1}{2\pi D_r} \log(1/a) + O(1). \quad (\text{S87})$$

For grids 161, 241, 321, 401, fits over $a = 0.05, 0.04, 0.03, 0.025, 0.02$ give slope ratios to $1/(2\pi)$ of 0.950, 0.979, 0.984, 0.985, with finest-grid $R^2 = 0.99996$

(Fig. S7). At fixed $\kappa = 1$, the reaction-limited benchmark $\langle T \rangle \sim 1/(\kappa\pi a^2)$ is recovered to within 0.83% over the tested radii.

S8.2. Three-dimensional effective radius

In three relative dimensions,

$$G(r, r') \sim \frac{1}{4\pi D_r |r - r'|} + H. \quad (\text{S88})$$

For a Doi sphere, matching the regular interior radial solution to the exterior capture field gives

$$a_{\text{eff}} = a \left(1 - \frac{\tanh \sqrt{\chi}}{\sqrt{\chi}} \right), \quad \chi = \frac{\kappa a^2}{D_r}, \quad (\text{S89})$$

and

$$k_{\text{Doi}}^{(3)} = 4\pi D_r a_{\text{eff}}. \quad (\text{S90})$$

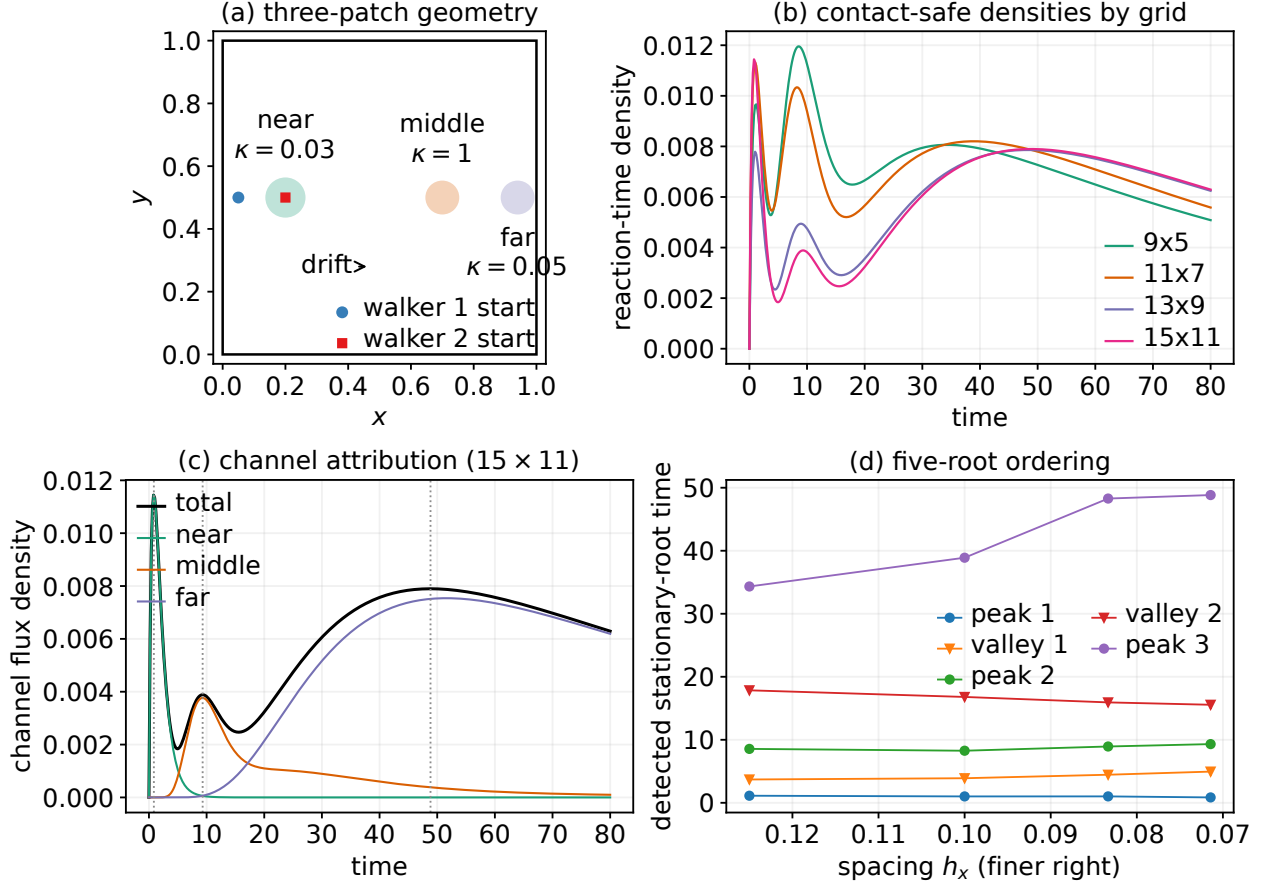


FIG. S6. Finite-radius bounded-domain multi-clock certificate. Three nonnegative midpoint-selected Doi patches generate three strict maxima on four boundary-node grids; the classifier resolves all three on the three finer grids and calls the coarsest a shoulder. Direct finite-matrix semigroup derivative evaluations detect five alternating sign-changing derivative roots, thereby establishing at least five positive-time roots, and the three maxima are attributed directly to the near, middle, and far channel fluxes. The repeated finite-grid pattern is based on underresolved binary patch masks, and the finite scan is not interval-exhaustive; it is not a cell-averaged continuum theorem or a converged trimodality phase boundary.

On a periodic volume V , the small-target mean has the form

$$\langle T \rangle = \frac{V}{4\pi D_r a_{\text{eff}}} + C_{\text{torus}} \frac{L^2}{D_r} + o(L^2/D_r). \quad (\text{S91})$$

The finite-volume correction is additive, so the clean test is the slope against $1/a_{\text{eff}}$, not equality to the leading term at finite radius.

The matrix-free periodic solver uses cell-averaged Doi spheres, conjugate gradients, and an FFT inverse of a shifted discrete Laplacian as preconditioner. Up to 4.83×10^6 states are treated without assembling the backward operator. Along the coupled fixed- χ radius/grid path, where a/h stays between about 7.4 and 7.6, the smallest-four-radius fit has a slope error 0.114% relative to $1/(4\pi)$. This is continuum-compatible finite-grid evidence, not a separated double-limit certification. At the separately refined fixed radius $a = 0.09$, the finest-grid-pair difference is 0.065%, and the largest relative linear

residual is 6.3×10^{-14} . At fixed κ ,

$$\langle T \rangle \sim \frac{V}{\kappa(4\pi a^3/3)} \quad (\text{S92})$$

is recovered at the smallest radius with 0.042% relative error. Figure S8 displays the fixed-radius convergence, coupled capacity path, and reaction-limited cross-check without conflating their distinct limiting statements.

S9. ADDITIONAL ANALYSES AND ANALYTICAL OBLIGATIONS

The finite controls distinguish transport-supplied clocks from their reaction-field weighting; neither transport alone nor heterogeneity alone is a general explanation. A visible time-domain fold is also not a spectral exceptional point: poles may be dark or cancelled in the selected observable.

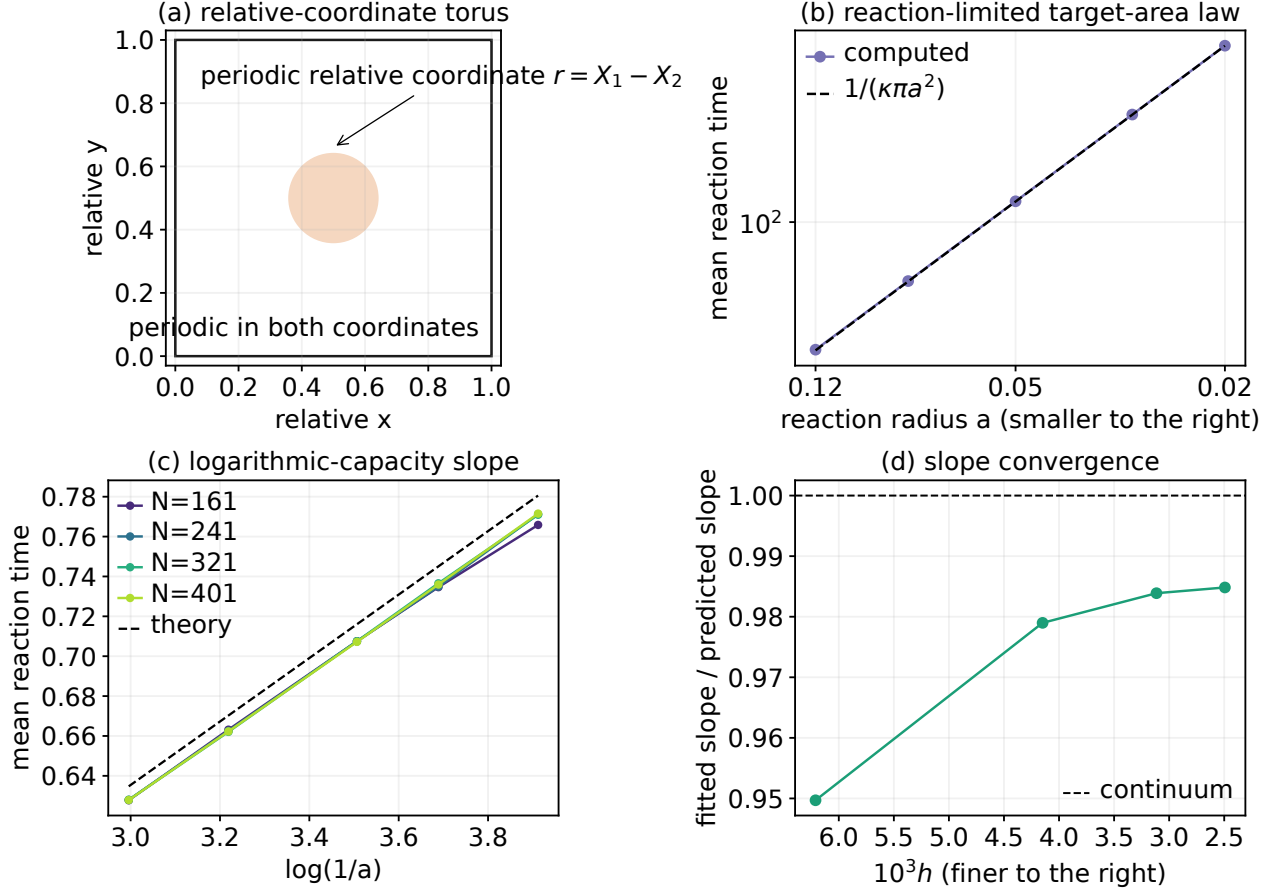


FIG. S7. Two-dimensional finite-radius calibration on a translation-invariant flat torus. Fixed- χ mean times approach the predicted logarithmic-capacity slope under grid refinement, while fixed- κ calculations recover the reaction-limited target-area law. These are relative-coordinate mean-time benchmarks, not centre-patterned modality calculations.

S9.1. Constructive multidimensional screening designs

The free-space GIG approximation also gives a constructive screening rule for more than two clocks. In physical dimension d , write $p = (d+3)/2$ and use a common $B > 0$. To place the isolated mode of channel j at a prescribed time m_j , choose

$$A_j = Bm_j^2 + pm_j, \quad w_j = \frac{g_j(m_j)^{-1}}{\sum_k g_k(m_k)^{-1}}. \quad (\text{S93})$$

The first identity solves the GIG mode equation exactly; the second equalizes the isolated weighted peak heights without an optimization. For the reference free-motion mapping $D_1 = D_2 = 1/2$, initial gap $\ell = 1$, zero relative drift $u = 0$, and centre speed $|v_c| = 0.1$, one has $B = 0.01$ and

$$|z_j - R_0| = \sqrt{Bm_j^2 + pm_j - \frac{1}{4}}. \quad (\text{S94})$$

More generally the distance is $\sqrt{4D_c(A_j - A_{\text{rel}})}$, so a real catalyst location requires $A_j \geq A_{\text{rel}} = \ell^2/(4D_r)$.

In the reference mapping this is $Bm_j^2 + pm_j \geq 1/4$, or $m_j \geq [-p + \sqrt{p^2 + B}]/(2B)$; the threshold ranges from 0.12492 in $d = 1$ to 0.071414 in $d = 4$, below every tested target $m_j \geq 1$.

For dimensions $d = 1, 2, 3, 4$ and target clocks (1, 10), (1, 10, 100), and (1, 10, 100, 1000), direct isolation of the analytical derivative roots gives respectively two, three, and four resolved modes in all 12 tested cases (Fig. S9). The finite logarithmic scan found $2m - 1$ alternating simple critical points for m channels in every case; an interval-certified exclusion of additional tangential roots is not claimed. The weakest peak-to-adjacent-valley ratio exceeds 1.9, and the largest scaled derivative residual is below 2×10^{-10} .

This result is a *free-space, narrow-patch GIG screening construction*. It is not a theorem for a bounded Doi problem. Reflections, finite-patch averaging, capacity renormalization, and the physical realization of the designed splitting weights—including compensation for the time-independent drift cross factors above—remain to be controlled before a three- or four-mode encounter claim can be made.

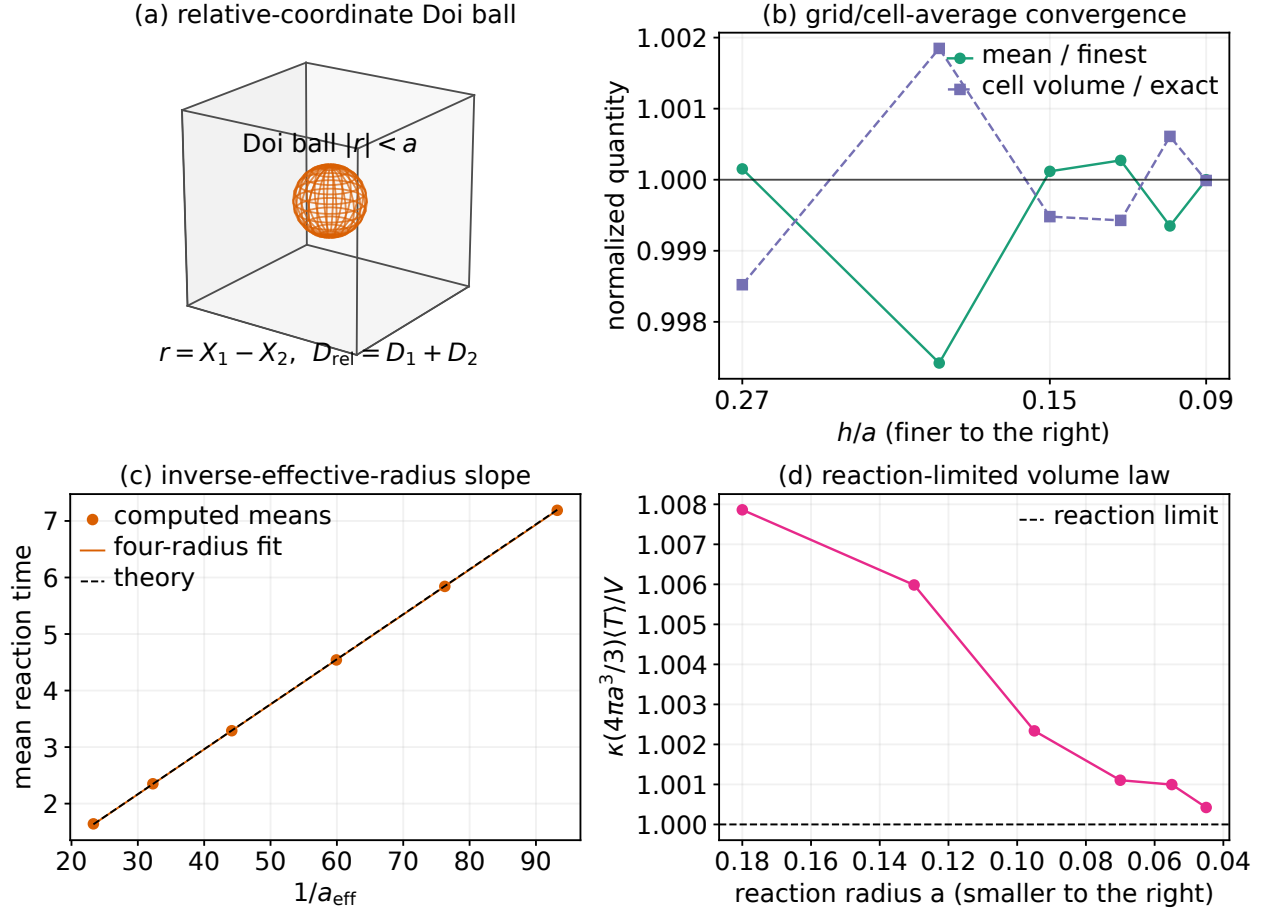


FIG. S8. Three-dimensional Doi capacity on a translation-invariant flat torus. Fixed-radius grid convergence, coupled-path inverse-effective-radius scaling, and the fixed- κ reaction-limited law test the relative-coordinate solver. The radius and grid limits of the shrinking-radius path are not separately extrapolated. Translation invariance is essential to the exact quotient; this calculation is not a six-dimensional centre-patterned modality result.

S9.2. Relation to prior work

The novelty boundary is deliberately narrow. Confined encounter distributions, Green and multi-target reductions, heterogeneous reactive locations, GIG hitting-time laws, capacity asymptotics, and multimodal first-passage densities are all established. Static heterogeneous bulk killing has already been optimized at fixed total strength, and partially reactive network elements have been rearranged at fixed total absorbance [7, 8]. Position-dependent Robin reactivity has also been optimized under a fixed integral constraint to maximize steady flux [9]. Fixed-total-reactivity spatial optimization is therefore not claimed here. Recent reactive capacitance theory treats arbitrary patch shapes [10]. In particular, the Brownian first-passage transition of Besga *et al.* already supplies a two-time-scale shape change and the merger criterion $f_t = f_{tt} = 0$ under a target-distance/initial-spread control [11]. It does not redistribute static Doi reactivity at fixed transport, initial law, and discrete

killing budget, which is the control studied here. The Giuggioli program already contains multiple peaks generated by periodic-domain bias, persistence, competing target routes, and small-world disorder, while resetting and inert heterogeneity reshape peak weights or the location of a first-passage mode [12–17], while Grebenkov showed that encounter-history-dependent surface activation can induce a mono-to-bimodal reaction-time change [18], and reactive-state gating can produce multimodal reaction times [19, 20].

We use those established propagator, Green, and channel ideas as a computational layer rather than claiming a new Green theory. The generalized Descartes rule and Fréchet–Duhamel calculus are likewise classical tools; their mathematical novelty is not claimed. The general increment is to join a necessary spectral-residue diagnostic to a fixed-budget modality gradient and test both on the same derivative-certified encounter-fold program. The spatial evidence then uses two separately declared matched families. M2D-E fixes its transport generator and discrete state-sum killing budget on each lattice to

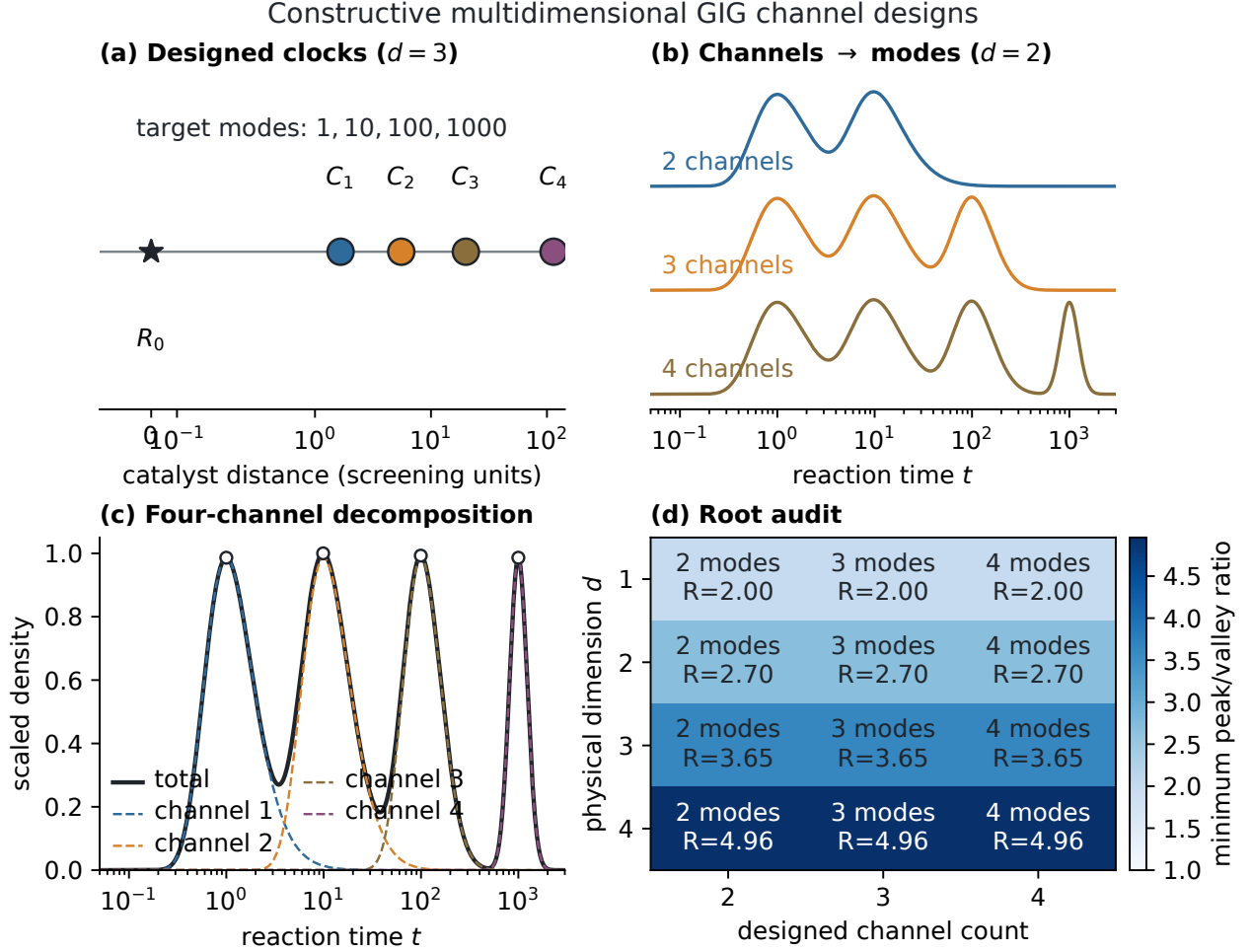


FIG. S9. Constructive multidimensional GIG channel designs. Prescribed clocks are mapped to catalyst distances and inverse-isolated-height weights. Analytical derivatives are numerically root-isolated for two, three, and four channels in $d = 1, \dots, 4$. The figure validates the declared GIG family only; it is not evidence for bounded finite-radius Doi trimodality.

test endpoint amplification under spatial redistribution. M2D-F fixes a different transport generator and its own per-lattice state-sum budget, then locates a density transition through $f_t = f_{tt} = 0$ with nondegeneracy and physical-path transversality. The supporting spatial results are the M2D-E five-grid endpoint audit with four resolved label contrasts, three nondegenerate but non-converged M2D-F finite-grid folds, and a bounded finite-grid three-channel mechanism certificate. They are not a first observation of multiple peaks or of a passage-time shape transition, a new Woodbury identity, or a continuum modality theorem. Static fixed-total-reactivity design, heterogeneous reaction, reactivity-generated bimodality, and the Green machinery are not claimed as new; the model-specific new object is the finite-grid density fold under the stated M2D-F matched experiment, with M2D-E providing a separate endpoint-amplification audit. The finite-state local-sensitivity calculus explains which budget-feasible perturbations can unfold such a boundary without asserting a continuum optimum.

S9.3. Limitations and analytical obligations

Several limitations are necessary to interpret the results correctly.

1. The GIG law has no uniform remainder over a mode-containing window and is not a global reflected-path expansion.
2. The inverse-free continuum Green identity requires operator-domain hypotheses. The implemented left-half-plane pole/residue and derivative validations are finite-dimensional; continuum meromorphic continuation is not established. Near the reflecting zero mode, the free-Green implementation is fail-closed and the stable zero-frequency quantities use the killed solve.
3. The spectral sign-variation corollary is necessary but not sufficient and is used only for finite reversible generators with grouped real decay rates.

- The primary-model spectra have far more sign changes than the observed roots, so the count is a falsification gate rather than a mode predictor. It does not apply unchanged to complex spectra, Jordan terms, or an infinite continuum expansion.
4. The budget-projected modality gradient is an infinitesimal interior-rate result. Its closed-form optimizer ignores finite-amplitude nonnegativity, box constraints, binary patch masks, and the nonsmooth motion of a sharp support. Those constraints require a separate quadratic, shape, or mixed-integer design problem.
 5. The physical fold is numerically located to the stated tolerances for one finite CTMC using exact finite-matrix derivative identities; it is not uniform in lattice spacing, radius, or system size. Its minority splitting probability is 8.95×10^{-6} . The corresponding absorption law is phase type; finite-state multimodality is not claimed as a new probability-law class.
 6. The M2D-E endpoint and M2D-F fold comparisons use moderate grids and an upwind Doi discretization. Cell Péclet numbers remain large, so the tested lattices are not a controlled refinement of the displayed continuum SDE. The three fold controls are nonmonotone, span 0.262, and move substantially under a product-control-volume budget measure. Their location is not grid-converged. The resolved endpoint contrast is operational: the full classifier is stated, but no experiment-specific detector or noise model derives its 3% scale. An independent Robin solver and Doi–Robin matching are not included. Even convergence of the density in C^2 would not by itself transfer a nondegenerate fold; the required target is joint C^3 convergence in time and control, including the parameter Duhamel derivatives.
 7. The bounded 2D model declares midpoint catalysis. The diffusion-decoupling weighted-coordinate control changes discrete masks and the quantitative densities even though the three tested morphology labels agree; no invariance with respect to η or equivalence of these physical sink models is claimed.
 8. Equal per-lattice discrete state-sum killing is one matching rule; it does not match every pathwise hazard, splitting weight, or mean lifetime. The separate M2D-C uniform-reactivity counterexample rules out necessity language.
 9. The interior far patch has clearance 0.07, but late trajectories can still feel the reflecting boundary. Enlarged-domain or periodic matched controls are needed before claiming boundary independence.
 10. The 2D and 3D capacity studies exploit translation invariance and concern mean reaction times. They do not prove full centre-patterned channel-density or modality convergence.
 11. Four-size scaling fits are descriptive. Peak separation alone does not prove a large-distance two-mode theorem; derivative-dominance bounds, weights, widths, and a controlled remainder are required.
 12. The M2D-T family establishes three classifier-resolved finite-grid trimodal cases and a fourth finite-grid density with three strict maxima but a classifier shoulder. Each grid has at least five detected sign-changing roots in the alternating pattern, but the finite scan is not an interval-exhaustive root certificate and establishes no cell-averaged continuum trimodal region, converged cusp, or general- d modality theorem. Every patch radius is below the longitudinal grid spacing, and the reactive-state counts are nonmonotone. The multi-channel GIG construction remains a free-space screening result.
- A continuum persistence theorem should decompose
- $$f_L = w_{E,L} g_{E,L} + w_{B,L} g_{B,L} + \varepsilon_L \quad (\text{S95})$$
- and control both channels and their first two derivatives in neighborhoods of the early peak, the late peak after centering and scaling, and the intervening valley. Nonvanishing limiting weights, nondegenerate channel maxima, and cross-channel derivative margins would then imply two persistent modes by stability of simple critical points. Transferring the fold itself additionally requires the joint C^3 control in Eq. (S63) and convergence of the budget-projected parameter derivative. Establishing these assumptions from the encounter generator, rather than fitting peak locations, is the next analytical step.

Appendix SA: Fold normal-form prominence

Let

$$F(t, \theta) = f_t(t, \theta) = a_* \Delta \theta + \frac{b_*}{2} \Delta t^2 + o(|\Delta \theta| + |\Delta t|^2). \quad (\text{SA1})$$

Set

$$c = \sqrt{-\frac{2a_*}{b_*} \Delta \theta}. \quad (\text{SA2})$$

At leading order the two new critical points are at $\Delta t_{\pm} = \pm c$. The density difference between the local maximum and minimum is the integral of F ,

$$\mathcal{P} = \left| \int_{-c}^c \left(a_* \Delta \theta + \frac{b_*}{2} x^2 \right) dx \right| \quad (\text{SA3})$$

$$= \frac{2|b_*|}{3} c^3 = \frac{2|b_*|}{3} \left| -\frac{2a_*}{b_*} \right|^{3/2} |\Delta \theta|^{3/2}, \quad (\text{SA4})$$

up to $o(|\Delta\theta|^{3/2})$. This gives the 3/2 exponent used in the independent CTMC validation.

Appendix SB: Finite-state mass and derivative identities

For a row generator T , initial row distribution α , and total killing vector $b = UK\mathbf{1} = -T\mathbf{1}$,

$$S(t) = \alpha e^{Tt}\mathbf{1}, \quad f(t) = \alpha e^{Tt}b. \quad (\text{SB1})$$

Therefore

$$-S'(t) = -\alpha e^{Tt}T\mathbf{1} = \alpha e^{Tt}b = f(t), \quad (\text{SB2})$$

which is the exact finite-state version of Eq. (S8). If the killed chain is transient,

$$\int_0^\infty \alpha e^{Tt}UK dt = \alpha(-T)^{-1}UK. \quad (\text{SB3})$$

Repeated time differentiation yields Eq. (S34). For $T = T(\theta)$, Duhamel's identity gives

$$\partial_\theta e^{Tt} = \int_0^t e^{T(t-s)}T_\theta e^{Ts} ds, \quad (\text{SB4})$$

which is equivalent to the augmented sensitivity propagation in Eq. (S70).

Appendix SC: Numerical evidence and provenance boundary

The numerical sources are the JSON and CSV artifacts accompanying this article. Each secondary calculation has a machine-readable manifest recording the command, source hashes, model specification, runtime, and output hashes. Before submission, these manifests must be regenerated from a clean tagged commit and aggregated by a transitive top-level manifest.

The accompanying Lean 4/mathlib package contains 140 theorem/lemma declarations, including 60 encounter-specific statements. Static source inspection finds no executable `sorry`, `admit`, or declared axiom. The preserved 2026-07-14 build and theorem-printing logs record a local build without errors, and the preserved theorem-by-theorem reports list only `propext`, `Classical.choice`, and `Quot.sound`. No independent Lean rebuild is claimed here. The encoded results cover the two-channel fold elimination and quadratic normal form, the two-hotspot inverse, scalar/componentwise affine-centre algebra and a one-dimensional absolute-value contact bound, the scalar relative-centre transform, the GIG stationary-point equation, the pure a^{d-2} capacity-power algebra without its geometric constant or a capacity theorem, finite exponential-mixture derivatives, and inverse-height design weights. In a general real inner-product design space, it also proves that the

one-budget projected gradient is tangent, preserves every feasible response, bounds every unit feasible response, and that its normalized nonzero direction attains the bound. Taking the inner product to be $\langle x, y \rangle_M = x^\top My$, with Riesz vectors $M^{-1}c$ and $M^{-1}g$, gives Eqs. (S45) and (S46). The prescribed-action results prove vanishing log derivative at the chosen time, not formal uniqueness or maximality of that stationary point. This formal build certifies only those algebraic implications. It does not certify PDE regularity, floating-point roots, grid or continuum limits, or the applicability of the encoded assumptions to the physical model.

Appendix SD: Proof of the fold-transfer theorem

Write $q = (t, \theta)$, $H(q) = (F_t, F_{tt})(q)$, $H_h(q) = (F_{h,t}, F_{h,tt})(q)$, $J = DH$. The four entries of J admit the augmented modulus $\hat{\omega}(r) = \max(\omega(r), \sqrt{2}\Lambda r)$: the entry F_{tt} is $\sqrt{2}\Lambda$ -Lipschitz (its gradient is $(F_{ttt}, F_{tt\theta})$, bounded entrywise by Λ), and $F_{t\theta}, F_{ttt}, F_{tt\theta}$ carry ω by hypothesis, so each entry of $J(q) - J(q')$ is at most $\hat{\omega}(\|q - q'\|)$, and $\hat{\omega}(r_0) \leq \hat{\omega}(r_\omega) \leq \sigma/16$ by the (nondecreasing) definition of r_ω . For a 2×2 matrix with entries bounded by $\hat{\omega}$, the operator norm is at most $2\hat{\omega}$; we use this factor explicitly. All norms are Euclidean and induced operator norms, and $B(q_*, r_0) \subseteq N$ by the definition of r_0 .

Step 0 (seed conditioning). At q_* , $F_{tt} = 0$, $|F_{ttt}| \geq \mu_3$, and $|F_{tt\theta}|, |F_{t\theta}| \leq \Lambda$, so

$$\sigma = \sigma_{\min}(J(q_*)) = \frac{|\det J(q_*)|}{\sigma_{\max}(J(q_*))} \geq \frac{|F_{t\theta}|\mu_3}{\sqrt{F_{t\theta}^2 + 2\Lambda^2}} \geq \frac{\mu_3\mu_\theta}{\sqrt{\mu_\theta^2 + 2\Lambda^2}},$$

using $\sigma_{\max} \leq \|J\|_F = \sqrt{F_{t\theta}^2 + F_{ttt}^2 + F_{tt\theta}^2}$ and monotonicity of $x \mapsto x/\sqrt{x^2 + 2\Lambda^2}$.

Step 1 (perturbation inventory). The defect matrix $DH_h - DH$ has entries bounded by ε_h (entry F_{tt}) and ε'_h (entries $F_{t\theta}, F_{ttt}, F_{tt\theta}$), so $\|DH_h - DH\| \leq \sqrt{\varepsilon_h^2 + 3\varepsilon_h'^2} \leq 2\bar{\varepsilon}_h$. By $\hat{\omega}(r_0) \leq \sigma/16$, $\|J(q) - J(q_*)\| \leq 2\hat{\omega}(r_0) \leq \sigma/8$ on $B(q_*, r_0)$.

Step 2 (simplified Newton). Assume $\bar{\varepsilon}_h < \delta_0 \leq \sigma/32$, so $2\bar{\varepsilon}_h \leq \sigma/16$ and $4\bar{\varepsilon}_h \leq \sigma/8$. At the seed, $\|DH_h(q_*) - J(q_*)\| \leq 2\bar{\varepsilon}_h \leq \sigma/16$, so by the Neumann series $\|DH_h(q_*)^{-1}\| \leq \frac{16}{15}\|J(q_*)^{-1}\| \leq \frac{8}{7}\|J(q_*)^{-1}\|$. Define $\Phi(q) = q - DH_h(q_*)^{-1}H_h(q)$. For $q \in B(q_*, r_0)$,

$$\|D\Phi(q)\| = \|DH_h(q_*)^{-1}(DH_h(q_*) - DH_h(q))\| \leq \frac{8}{7} \cdot \frac{1}{\sigma} (2\hat{\omega}(r_0) + 4\bar{\varepsilon}_h)$$

since $\|DH_h(q_*) - DH_h(q)\| \leq \|(DH_h - DH)(q_*)\| + \|DH(q_*) - DH(q)\| + \|(DH - DH_h)(q)\| \leq 2\bar{\varepsilon}_h + 2\hat{\omega}(r_0) + 2\bar{\varepsilon}_h$. Also $\|\Phi(q_*) - q_*\| = \|DH_h(q_*)^{-1}H_h(q_*)\| \leq \alpha$ with $\alpha := \frac{8\sqrt{2}}{7}\|J(q_*)^{-1}\|\varepsilon_h$, because $\|H_h(q_*)\| = \|(H_h - H)(q_*)\| \leq \sqrt{2}\varepsilon_h$. The fourth term of δ_0 gives $\frac{7}{5}\alpha = \frac{8\sqrt{2}}{5\sigma}\varepsilon_h \leq r_0$, so Φ maps $B(q_*, \frac{7}{5}\alpha) \subseteq B(q_*, r_0)$ into itself ($\alpha + \frac{2}{7} \cdot \frac{7}{5}\alpha = \frac{7}{5}\alpha$) and is a $\frac{7}{5}$ -contraction there. The

Banach fixed point q_h satisfies $H_h(q_h) = 0$ and

$$\|q_h - q_*\| \leq \frac{\alpha}{1 - \frac{2}{7}} = \frac{7}{5}\alpha = \frac{8\sqrt{2}}{5} \|J(q_*)^{-1}\| \varepsilon_h \leq 2\sqrt{2} \|J(q_*)^{-1}\| \varepsilon_h$$

Step 3 (margins and uniqueness in the fixed ball). The margin hypotheses hold on all of N , so directly $|F_{h,ttt}| \geq \mu_3 - \bar{\varepsilon}_h > \mu_3/2 > 0$ and $|F_{h,t\theta}| \geq \mu_\theta - \bar{\varepsilon}_h > \mu_\theta/2 > 0$ everywhere on N , in particular at q_h : the fold is nondegenerate with the stated margins. If $H_h(a) = H_h(b) = 0$ with $a, b \in B(q_*, r_0)$, the averaged Jacobian $M = \int_0^1 DH_h(a + s(b - a)) ds$ obeys $M(b - a) = 0$ and

$$\|M - J(q_*)\| \leq 2\hat{\omega}(r_0) + 2\bar{\varepsilon}_h \leq \frac{\sigma}{8} + \frac{\sigma}{16} = \frac{3\sigma}{16} < \sigma,$$

so M is invertible and $a = b$.

Converse. Under the ladder hypotheses, apply Steps 0–3 with the roles of F and F_h exchanged, seeded at q_h : Step 0 gives $\sigma_h = 1/\|J_h(q_h)^{-1}\| \geq \mu_3\mu_\theta/\sqrt{\mu_\theta^2 + 2\Lambda^2}$ uniformly in h ; the common augmented modulus $\hat{\omega}$ gives a common r_0 , and $\text{dist}(q_h, \partial N) \geq d$ keeps $B(q_h, \min(r_0, d)) \subseteq N$; uniform jet convergence makes the defects $\varepsilon_h, \bar{\varepsilon}_h \rightarrow 0$, so the threshold δ_0 (with r_0 replaced by $\min(r_0, d)$) is met for fine h , and F acquires a unique nondegenerate fold $\bar{q}$ in the ball with $\|\bar{q} - q_h\| \leq 2\sqrt{2}\|J_h(q_h)^{-1}\|\varepsilon_h \leq 2\sqrt{2}\sqrt{\mu_\theta^2 + 2\Lambda^2}(\mu_3\mu_\theta)^{-1}\varepsilon_h$. $\square$

Appendix SE: Continuum-bridge data

Table SV lists every accepted root of the cell-averaged square-cell refinement ladder (main text Sec. V (Con-

tinuum bridge)), with normalized margins $f_{ttt}t_c^3/f$ and $f_{t\theta}t_c/f$ and the larger of the two normalized first- and second-derivative residuals. The pre-asymptotic branch structure at $n \leq 13$ is discussed in the main text. Budget θ -invariance holds to 2×10^{-12} relative at every level; free-generator mass balance to 3×10^{-16} .

The off-lattice runs (main text Sec. V (Continuum bridge)) use 2×10^7 walkers per control value at $dt = 2 \times 10^{-4}$, $t_{\max} = 120$, seeded substreams per chunk. Second-maximum significances are Poisson z -values of the smoothed-histogram prominence under a five-sigma prominence rule; this diagnostic is distinct from the repository double-peak classifier. Per- θ event counts in the diagnostic window $t \in [10, 30]$: 8.0×10^4 ($\theta = 0.05$), 9.9×10^4 (0.12), 1.8×10^5 (0.30), 5.1×10^5 (0.60), 1.4×10^6 (0.85). The runs were executed on the Isambard-AI facility; job-level provenance (grid sizes, runtimes, and artifact hashes) is recorded in the accompanying data files.

DATA AVAILABILITY

All source code, numerical tables, vector figures, classifier specifications, and machine-readable provenance manifests used in this study are contained in the accompanying research repository. A public archival DOI will be supplied at submission.

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TABLE SV. Refinement-ladder roots (UW = upwind, SG = Scharfetter–Gummel).

n	scheme	θ_c	t_c	$f_{ttt}t_c^3/f$	$f_{t\theta}t_c/f$	residual
9	SG	0.2148081	18.60708	-27.48	0.84	1.6×10^{-13}
9	UW	-0.1836682	10.38672	-43.04	-87.87	1.2×10^{-13}
13	SG	-0.1178265	19.32734	-31.59	16.24	7.8×10^{-14}
13	UW	-0.1757642	10.57799	-56.23	-98.90	2.1×10^{-12}
17	SG	-0.1385771	18.29303	-25.94	24.58	8.5×10^{-13}
17	UW	-0.0431218	17.95201	-26.97	3.57	6.8×10^{-15}
21	SG	-0.1458343	17.79574	-22.93	31.90	3.0×10^{-13}
21	UW	-0.1000190	18.27339	-27.05	9.63	3.7×10^{-13}
25	SG	-0.1507108	17.44176	-20.12	35.10	3.9×10^{-13}
25	UW	-0.1181117	18.30113	-26.86	14.32	3.3×10^{-13}
29	SG	-0.1552607	16.97706	-16.02	35.65	1.1×10^{-13}
29	UW	-0.1297125	18.22777	-25.20	18.59	1.1×10^{-12}
33	SG	-0.1578014	16.81058	-13.30	38.07	1.1×10^{-12}
33	UW	-0.1370191	18.17374	-23.91	22.19	2.4×10^{-12}

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